DEPARTMENT OF CHEMICAL ENGINEERING

Course Book for

B. Tech. in Chemical Engineering

For

Academic Year
2024- 2025

Visvesvaraya National Institute of Technology

Nagpur 440 010 (India)

Institute Vision Statement

To contribute effectively to the National and International endeavor of producing quality human resource of world class standard by developing a sustainable technical education system to meet the changing technological needs of the Country and the World incorporating relevant social concerns and to build an environment to create and propagate innovative technologies for the economic development of the Nation.

Institute Mission Statement

The mission of VNIT is to achieve high standards of excellence in generating and propagating knowledge in engineering and allied disciplines. VNIT is committed to providing an education that combines rigorous academics with joy of discovery. The Institute encourages its community to engage in a dialogue with society to be able to effectively contribute for the betterment of humankind.

List of faculty Members

Sr No	Faculty Name	Areas of specialization
1	Dr. K. L. Wasewar	Separation Processes, Ionic liquids
2	Dr. Sachin. A. Mandavagane	Bio-fertilizer, adsorption, bio-adsorbent
3	Dr. M. N. Varma	Reaction Engineering, Bio-fuel, Supercritical fluids
4	Dr. S. S. Sonawane	Nanotechnology, nano-fluids, polymer composite
5	Dr. A. S. Chaurasia	Pyrolysis, gasification, CFD
6	Dr. Diwakar Z. Shende	Separation processes, mass transfer
7	Dr. Ajit Rathod	Membrane separation, catalysis, heat transfer
8	Dr. Pradeep Dhamole	Bio-fuel, ionic liquids, separation processes
9	Dr. R. Methekar	Process control, Fuel cell, batteries
10	Dr. Vijayakumar R. P.	Polymer nano-composit, carbon nano-tube
11	Dr. C. Ravikumar	Bio-nanotechnology, hydrodynamics
12	Dr. A. P. Gaikwad	Bio-fuel, Catalysis
13	Dr. Shyam M. Kodape	Membrane Bioreactor, Industrial Waste Treatment, Waste Valorization, Green Synthesiss
14	Dr. S. P. Tajane	Bio-fertilizer, mechanical operations
15	Dr. P. Wanjari	Nano-technology, molecular dynamic simulation
16	Dr. A. Kannan	Bio-nano film, separation processes
17	Dr Arijit Das	Separation processes, extraction, ceramic membrane

Credit System at VNIT:

Education at the Institute is organized around the semester-based credit system of study. The prominent features of the credit system are a process of continuous evaluation of a student's performance / progress and flexibility to allow a student to progress at an optimum pace suited to his/her ability or convenience, subject to fulfilling minimum requirements for continuation. A student's performance/progress is measured by the number of credits he/she has earned, i.e. completed satisfactorily. Based on the course credits and grades obtained by the student, grade point average is calculated. A minimum number of credits and a minimum grade point average must be acquired by a student in order to qualify for the degree.

Course credits assignment

Each course, except a few special courses, has certain number of credits assigned to it depending on lecture, tutorial and laboratory contact hours in a week.

For Lectures and Tutorials: One lecture hour per week per semester is assigned one credit and

For Practical/ Laboratory/ Studio: One hour per week per semester is assigned half credit.

Example: Course XXXXXX with (3-0-2) as (L-T-P) structure, i.e. 3 hr Lectures + 0 hr Tutorial + 2 hr Practical per week, will have (3x1 + 0x1 + 2x0.5 =) 4 credits.

Grading System

The grading reflects a student's own proficiency in the course. While relative standing of the student is clearly indicated by his/her grades, the process of awarding grades is based on fitting performance of the class to some statistical distribution. The course coordinator and associated faculty members for a course formulate appropriate procedure to award grades. These grades are reflective of the student's performance vis-à-vis instructor's expectation. If a student is declared pass in a subject, then he/she gets the credits associated with that subject.

Depending on marks scored in a subject, a student is given a Grade. Each grade has got certain grade points as follows:

Grade	Grade points	Description
AA	10	Outstanding
AB	9	Excellent
BB	8	Very good
BC	7	Good
CC	6	Average
CD	5	Below average
DD	4	Marginal (Pass Grade)
FF	0	Poor (Fail) /Unsatisfactory / Absence from end-sem exam
NP	-	Audit pass

NF	-	Audit fail
SS	-	Satisfactory performance in zero credit core course
ZZ	-	Unsatisfactory performance in zero credit core course
W	-	Insufficient attendance

Performance Evaluation

The performance of a student is evaluated in terms of two indices, viz, the Semester Grade Point Average (SGPA) which is the Grade Point Average for a semester and Cumulative Grade Point Average (CGPA) which is the Grade Point Average for all the completed semesters at any point in time. CGPA is rounded up to second decimal.

The Earned Credits (ECR) are defined as the sum of course credits for courses in which students have been awarded grades between AA to DD. Grades obtained in the audit courses are not counted for computation of grade point average.

Earned Grade Points in a semester (EGP) = **Σ** (Course credits x Grade point) for courses in which AA- DD grade has been obtained

SGPA = EGP / Σ (Course credits) for courses registered in a semester in which AA- FF grades are awarded

CGPA= EGP / Σ(Course credits) for courses passed in all completed semesters in which AA- DD grades are awarded

Attendance Rules

1. All students must attend every class and 100% attendance is expected from the students. However, in consideration of the constraints/ unavoidable circumstances, the attendance can be relaxed by course coordinator only to the extent of not more than 25%. Every student must attend minimum of 75% of the classes actually held for that course.
2. A student with less than 75% attendance in a course during the semester, will be awarded W grade. Such a student will not be eligible to appear for the end semester and re-examination of that course. Even if such a student happens to appear for these examinations, then, answer books of such students will not be evaluated.
3. A student with W grade is not eligible to appear for end semester examination, reexamination & summer term.

Overall Credits Requirement for Award of Degree

SN	Category of Course	Symbol	Credit Requirement B. Tech. (4-Year)
Program Core			
1	Basic Sciences (BS)	BS	18
2	Engineering Arts & Sciences (ES)	ES	20
3	Humanities	HU/ HM*	05
4	Departmental core	DC	82
Program Elective			
1	Departmental Elective	DE	33-45
2	Humanities & Management	HM	0-6
3	Open Course	OC	0-6
Total requirement: BS + ES + DC+ DE + HM + OC =			**170**
Minimum Cumulative Grade Point Average required for the award of degree			**4.00**

AU-Audit, BS – Basic Sciences, DC- Department Core, DE-Department Elective, ES-Engineering Sciences, HM-Humanities, OC-Open Course
Any undergraduate student of VNIT must complete 170 credits (as mentioned above) to be eligible for degree.

Program Outcomes (Department Specific) for B.Tech in Chemical Engineering

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Curriculum of the courses of study
First Year (Semester I & II)
Courses to Register in First Year B.Tech.
(*Sections R, S, T, U, L*)

I Semester				
Code	**Course**	**Type**	**L-T-P**	**Credits**
AML151	Engineering Mechanics	ES	3-1-0	4
AMP151	Engineering Mechanics Laboratory	ES	0-0-2	1
HUL101	Communication Skills	HM	2-0-2	3
MAL101	Mathematics – I	BS	3-1-0	4
MEL101	Engineering Drawing	ES	3-0-0	3
MEP101	Engineering Drawing Practical	ES	0-0-2	1
PHL101	Physics	BS	3-1-0	4
PHP101	Physics Laboratory	BS	0-0-2	1
SAP101	Health Information and Sports –Part I	AU	0-0-2	0
	Total Credits			21
II Semester				
Code	**Course**	**Type**	**L-T-P**	**Credits**
CHL101	Chemistry	BS	3-1-0	4
CHP101	Chemistry Laboratory	BS	0-0-2	1
CSL101	Computer Programming	ES	3-0-2	4
EEL101	Electrical Engineering	ES	3-1-0	4
EEP101	Electrical Engineering Laboratory	ES	0-0-2	1
HUL102	Social Science	HM	2-0-0	2
MAL102	Mathematics – II	BS	3-1-0	4
MEP102	Workshop	ES	0-0-4	2
SAP102	Health Information and Sports –Part II	AU	0-0-2	0
	Total Credits			22

L-T-P = 3-1-0 Means, Three Theory Hrs + One Tutorial hrs + Zero Labs or Practical Hrs per Week. L-T-P = 0-0-2 Means, Two Hrs of Lab or Practical per Week

Courses to Register in First Year B.Tech.
(*Sections W, X, Y, Z, N*)

I Semester				
Code	**Course**	**Type**	**L-T-P**	**Credits**
CHL101	Chemistry	BS	3-1-0	4
CHP101	Chemistry Laboratory	BS	0-0-2	1
CSL101	Computer Programming	ES	3-0-2	4
EEL101	Electrical Engineering	ES	3-1-0	4
EEP101	Electrical Engineering Laboratory	ES	0-0-2	1
HUL102	Social Science	HM	2-0-0	2
MAL101	Mathematics – I	BS	3-1-0	4
MEP102	Workshop	ES	0-0-4	2
SAP102	Health Information and Sports –Part I	AU	0-0-2	0
	Total Credits			22
II Semester				
Code	**Course**	**Type**	**L-T-P**	**Credits**
AML151	Engineering Mechanics	ES	3-1-0	4
AMP151	Engineering Mechanics Laboratory	ES	0-0-2	1
HUL101	Communication Skills	HM	2-0-2	3
MAL102	Mathematics – II	BS	3-1-0	4
MEL101	Engineering Drawing	ES	3-0-0	3
MEP101	Engineering Drawing Practical	ES	0-0-2	1
PHL101	Physics	BS	3-1-0	4
PHP101	Physics Laboratory	BS	0-0-2	1
SAP101	Health Information and Sports –Part II	AU	0-0-2	0
	Total Credits			21

L-T-P = 3-1-0 Means, Three Theory Hrs + One Tutorial hrs + Zero Labs or Practical Hrs per Week.
L-T-P = 0-0-2 Means, Two Hrs of Lab or Practical per Week

3rd Semester B.Tech. Chemical Engg.

Code	Course Title	Type	Structure (L-T-P)	Credit
CML201	Chemical Process Calculations	DC	3-1-0	4
CML202	Fluid Mechanics	DC	3-1-0	4
CML203	Mechanical Operations	DC	3-0-0	3
CML204	Chemical Engg. Thermodynamics I	DC	3-0-0	3
CHL263	Organic Chemistry and Synthesis	DC	3-0-0	3
CHP263	Organic Chemistry and Synthesis Lab.	DC	0-0-2	1
CMP204	Technical Analysis Laboratory	DC	0-0-2	1
	Elective (Anyone)			
CML231	Material Science and Engineering	DE	3-0-0	3
CML392	Corrosion Engineering	DE	3-0-0	3
	Total Credits	DC+DE=19+3=22		

4th Semester B.Tech. Chemical Engg.

Code	Course Title	Type	Structure (L-T-P)	Credit
CML221	Mass Transfer I	DC	3-1-0	4
CML222	Heat Transfer	DC	3-1-0	4
CML223	Chemical Reaction Engineering -I	DC	3-1-0	4
CML224	Chemical Engineering Mathematics	DC	3-0-0	3
CML225	Chemical Engineering Thermodynamics -II	DC	3-1-0	4
CMP202	Fluid Mechanics Lab	DC	0-0-2	1
CMP203	Mechanical Operations Lab	DC	0-0-2	1
CMP322	Heat Transfer Lab	DC	0-0-2	1
	Elective (Anyone)			
CML233	Environmental Engineering	DE	3-0-0	3
CML234	Industrial Waste Treatment	DE	3-0-0	3
	Total Credits	DC+DE=22+3=25		

5^th Semester B.Tech. Chemical Engg.

Code	**Course Title**	**Type**	**Structure (L-T-P)**	**Credit**
CML301	Mass Transfer II	DC	3-0-0	3
CML302	Chem. Process Modeling & Simulation	DC	3-0-0	3
CML303	Process Control and Instrumentation	DC	3-0-0	3
CML304	Chemical Reaction Engineering II	DC	3-0-0	3
CML353	Chemical Process Equipment Design	DC	3-0-0	3
CMP304	Chemical Reaction Engg. laboratory	DC	0-0-2	1
CMP321	Mass Transfer laboratory	DC	0-0-2	1
CMP453	Design Lab	DC	0-0-2	1
	Elective (Anyone)			
CML232	Intro. to Computing Soft. for Chem. Engrs	DE	3-0-0	3
CML383	Advance Heat Transfer	DE	3-0-0	3
CML524	Polymer Science and Engg.	DE	3-0-0	3
	Total Credits	DC+DE=18+3=21		

6^th Semester B.Tech. Chemical Engg.

Code	**Course Title**	**Type**	**Structure (L-T-P)**	**Credit**
CML351	Chemical Technology	DC	3-0-0	3
CML352	Transport Phenomena	DC	3-1-0	4
CML354	Energy and Environment	DC	3-0-0	3
CMP302	Chem. Process Modeling and Simulation Lab	DC	0-0-2	1
CMP303	Process Control and Instrumentation Lab	DC	0-0-2	1
CML403	Plant Design and Economics	DC	3-0-0	3
CMP401	Mass Transfer laboratory II	DC	0-0-2	1
CMP403	Design Lab II	DC	0-0-2	1
	Elective (Any two)			
CML393	Innovative Design	DE	2-0-2	3
CML429	Nanotechnology	DE	3-0-0	3
CML424	Petroleum Refinery Engineering	DE	3-0-0	3
CML423	Optimization Techniques	DE	3-0-0	3
	Total Credits	DC+DE=17+6=23		

7th Semester B.Tech. Chemical Engg.

Code	Course Title	Type	Structure (L-T-P)	C
CMD401	Project Phase-I	DC		3
	Elective (Any five)			
CML402	Molecular Simulation and Informatics	DE	3-0-0	3
CML537	Sustainable Engineering	DE	3-0-0	3
CML425	Membrane Technology	DE	3-0-0	3
CML426	Polymer Processing	DE	3-0-0	3
CML427	Advanced Separation Process	DE	3-0-0	3
CML428	CFD for Chemical Engineers	DE	3-0-0	3
CML481	Instrumental Analytical Techniques	DE	3-0-0	3
CML529	Optimization Techniques in Process Design	DE	3-0-0	3
CMLXXX	MOOC-I	DE	3-0-0	3
OC/HM	OC/HM	DE	3-0-0	3
	Total Credits	DC+DE=3+15=18		

8th Semester B.Tech. Chemical Engg.

Code	Course Title	Type	Structure (L-T-P)	Credit
CMD402	Project Phase-II	DC		3
	Elective (Any five)			
CML386	Biotechnology and Biochemical Engg.	DE	3-0-0	3
CML422	Plant Utility	DE	3-0-0	3
CML431	Enterpreneurship Development	DE	3-0-0	3
CML468	Ore and Mineral Processing	DE	3-0-0	3
CML530	Process Intensification	DE	3-0-0	3
CML342	Safety and Risk Analysis	DE	3-0-0	3
CML4**	MOOC-2	DE	3-0-0	3
CML4**	MOOC-3	DE	3-0-0	3
CMP4**	Internship	DE	3-0-0	3
OC/HM	OC/HM	OC/HM	3-0-0	3
	Total Credits	DC+DE=3+15=18		

Course Code : CML 201
Course Title : Chemical Process Calculations
Course Credits (LTP) : 4 Credits (3-1-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Brief Course Description:

This subject deals with the principles of stoichiometry to formulate and solve material and energy balance problems in processes with and without chemical reactions.

Course Objectives:

1) To understand the fundamental concepts and calculations of process calculation.
2) To understand the material balance in various unit processes and unit operations.
3) To understand the energy balance related to various process equipments.
4) To understand the various heats and their calculations related to chemical reactions including combustion.

Course Outcomes:

CO1: Ability to sort a process into different unit operations and processes and apply basic chemical calculations to it

CO2: Ability to apply material balances to different physical steps in a process

CO3: Ability to quantify heat and energy changes accompanying unit operations and unit processes

CO4: Ability to effectively combine the knowledge of material and energy calculations for processes

Mapping Course Outcomes to Program Outcomes:

CO	PO											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3		2						
CO 2	3	3	3			1	2					
CO 3	3	3	3	2			2					
CO 4	3	3	3	2			2			2		

Expanded Course Description:

Fundamental concepts, gas relationship, molarity, molality, normality, partial pressure, pure component volume and the related calculations. Humidity and saturation and their applications fundamental concepts of material balance.

Material balance in various unit processes and unit operations.

Material balance with chemical reactions Energy balance related to various process equipment.
Calculation of standard heat of reaction from the heat of formation and heat of combustion, thermochemistry, energy balance in various unit operations, the heat of solutions, the heat of neutralization etc.
Fuels and combustion calculation, proximate and ultimate analysis, adiabatic reaction temperature, air to fuel ratio, complex processes calculation.

Text Books:

1) Narayanan K V and Lakshmikutty B, Stoichiometry and Process Calculations, Prentice Hall of India Pvt Ltd, New Delhi 2006
2) Himmelblau D.M.; Basic Principles and Calculations in Chemical Engineering, 6th Edition, Prentice Hall of India Ltd.
3) Bhatt B.I. and Thakore S.B. Stoichiometry, 5th Edition, McGraw Hill Education.
4) Felder R.M., Rousseau R.W. and Bullard L.G.; Elementary Principles of Chemical Processess. 4th Edition, Wiley,

Evaluation Scheme:

- Class Tests and Assignments: 20%
- Mid-Semester Exam (1.5 Hours): 30%
- End-Semester Exam (3 Hours): 50%

Course Code : CML 202
Course Title : Fluid Mechanics
Course Credits (L-T-P) : 4 Credits (3-1-0)
Pre-requisites : None
Overlapping with : None

Course Objectives

1) To understand the fundamental properties, laws and their applications related to fluids.
2) To understand the fluid kinetics and fluid dynamics.
3) To understand the fluid flow through various sections.
4) To understand the flow over immersed bodies.
5) To understand the Boundary Layer Theory.
6) To understand the various types of pumps and their working principles.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. a recognition of the need for, and an ability to engage in lifelong learning

knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	h	i
1	√	√				√		√	
2	√	√				√	√	√	
3	√	√				√	√	√	
4	√	√				√		√	
5	√	√						√	
6	√	√				√		√	

Course Content (CO wise):

Properties of Fluid: Pressure, density, specific weight, viscosity, dynamic and kinemetic viscosity, Newton’s law of viscosity and its applications.

Fluid Statics: Pascal's Law and Hydrostatic equation, absolute and gauge pressures - pressure measurements by manometers and pressure gauges. – Forces on plane and curved surfaces

Fluid kinetics: Description of Fluid flow, Lagrangian and Eulerian approach One dimensional flow approximation, Types of fluid Flows: Steady and unsteady, Uniform and non-uniform, control volume concept, Reynolds transport theorem, Continuity equation, Velocity and acceleration of fluid particle, stream line, streak line, path line, velocity potential function,

Fluid Dynamics: Momentum theorem and its application. Euler's equation, Bernoulli's equation for incompressible fluid flow, Engineering applications of energy equation,

Pitot – static probe, Current meters, Venturimeter, Orificemeter, Rotameter, Nozzlemeter, Notches & weirs.

Flow Through Pipes: Critical Reynold's number, velocity distribution in pipes, friction factor, Moody's chart, Laminar flow through pipe, Hagen-Poiseulli's equation, turbulent flow through pipe, Hydraulic gradient line and Total energy line. Minor head losses in pipes. Pipe Networking Transmission of power through pipe.

Flow Over Immersed Bodies: Drag and lift, Types of drag force, Drag on sphere, Cylinder and airfoil; Circulation and Lift on a cylinder and airfoil; Magnus effect

Boundary Layer Theory: Development of Boundary layer over flat plate and pipe, boundary layer thickness

Pumps: definition and classifications - Centrifugal pump: classifications, working principles, specific speed, efficiency and performance curves - Reciprocating pump: classification, working principles, indicator diagram, work saved by air vessels and performance curves - cavitations in pumps - rotary pumps: working principles of gear and vane pumps

References:

1) Munson BR, Young D F and Okiishi T H , 'Fundamentals of Fluid Mechnics', 5th Edition, John Wiley & Sons
2) Gupta Santosh & Gupta Vijay, 'Fluid Mechanics and its applications', New Age International Publishers.
3) Munson BR, Young D F and Okiishi T H , ' Fundamentals of Fluid Mechnics', 5th Edition, John Wiley & Sons
4) Warren McCabe, Julian Smith, Peter Harriott, Unit Operations of Chemical Engineering, McGraw Hill International Edition, 2005

Course Code : CML203
Course Title : Mechanical Operation
Course Credits (L-T-P) : 3 credits(3-0-0)
Pre-requisites : None
Overlaps with : None

Course Objectives:
1. To impart the basic concepts of mechanical operations
2. To develop understanding about size analysis, size reduction and solid handling
3. Understand mechanical separation methods such as filtration, sedimentation, transportation of solids etc and associated equipments used for achieving these methods
4. The students are exposed to basic theory, calculations and machinery involved in various solid handling operations

Course Outcomes:
1. Able to describe properties of particulate solid and equipment for size reduction
2. Understand particle size analysis
3. Understand Filtration and centrifugal separation
4. Understand Motion of particles through fluid
5. Understand Agitation and mixing
6. Know different conveying system for handling of solids

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
CO1	**H**			**H**			**H**	**M**	
CO 2	**H**		**H**	**H**			**H**	**M**	
CO 3	**H**		**H**	**H**	**L**	**H**	**H**	**H**	**M**
CO 4	**H**		**H**	**H**			**H**	**H**	
CO 5	**H**	**L**		**H**	**H**	**H**	**H**	**H**	**H**
CO6	**H**	**L**		**H**	**H**	**H**	**H**	**H**	**H**

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams

g. The broad education necessary to understand the impact of engineering solutions in a global and societal context

h. A recognition of the need for, and an ability to engage in lifelong learning

i. A knowledge of contemporary issues

Course Content: This course deals with properties and characterization of solids and mechanical separation methods such as screening, filtration, sedimentation, transportation of solids, agitation etc and associated equipments used for achieving these methods.

CO 1: Properties of particulate solid and equipment for size reduction: Surface area distribution of powders, size reduction and separation, crushing, grinding equipments and their characteristics, open and close circuit grinding.

CO 2: Size Analysis: Particle size distribution, Screen analysis, mechanical classifiers classification.

CO 3: Filtration and centrifugal separation: Principles of filtration and theory, filtration equipments and their characteristics, pressure and vacuum filters, compressible and non compressible cake and their effect on filtration rate, centrifugal separation equipments and their principles of operation as well as the characteristics, optimum filtration cycle, membrane filtration.

CO 4: Motion of particles through fluid: Drag coefficient, free settling and hindered settling, gravity settlers, sedimentation theory and principle of operation. Batch and continuous thickeners as well as the design procedures, sedimenting centrifuges.

CO 5: Agitation and mixing: Introduction to agitation and mixing of solids and liquids fundamentals, mixing and agitation equipments and their operational characteristics, power consumption in mixing and agitation, different types of agitators and their selection criteria.

CO 6: Handling of solids: Storage and conveying of solids, bins, hoppers, silos and their operational characteristics, Loading and unloading of solids, different types of conveyors and elevators for solid materials. Dust collectors, cyclone separators, electrostatic precipitators, bag filters, operational characteristics of these and other similar dust separators.

Text Books:

1) Chattopadhyay O.P., Unit Operations of Chemical Engineering, Vol. 1 & 2, Khanna Publications, New Delhi, 1996.

2) Badger & Banchero Introduction to Chemical Enginerring Mc-Graw- Hill Education

3) G.G. Brown Unit Operation John Willey

4) Hiremath R.S & Kulkarni A.P.,. Mechanical Operations Vol I Everest Publication

Reference Books:

1) McCabe W.L., Smith J.C. and Harriot P., Unit Operations of Chemical Engineering McGraw Hill, New York 2001. 6th Edition

2) Coulson J. M. and Richardson J.F; Chemical Engineering Vol. 1& 2 Publishers: Butter worth – Heinemann Ltd. 2001-2002.

3) Christie J. GeanKoplis Transport processes & Unit Operation Prentice hall International

Course Code : CML204
Course Title : Chemical Engineering Thermodynamics-I
Course Credits (L-T-P) : Credits 3 (3-0-0)
Pre-requisites : Physical Chemistry and General Metallurgy
Overlaps with : None

Course Objectives:

1) To provide an introduction to chemical engineering thermodynamics as a fundamental component of chemical engineering,
2) To acquire the students with the knowledge for thermodynamic treatment of pure fluids and solutions
3) To understand the thermodynamics of phase equilibria and chemical reaction equilibria.
4) To provide the knowledge of working of Carnot cycle, refrigeration, compressors and nozzles.

Course Outcomes: At the end of this course student will

1. Understand the laws of thermodynamics.
2. be able to calculate changes in U, H, and S for ideal gases, and also for nonideal gases.
3. understand the utility of fugacity as a transformation of the chemical potential.
4. understand the criteria for chemical reaction equilibria.
5. Be able to understand the working of Carnot cycle, refrigeration, compressors and nozzles.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	h	i
CO1	√	√						√	
CO2	√	√	√		√	√		√	

CO3	√	√	√					√	
CO4	√	√		√				√	

Course Content:

It helps chemical engineer to evaluate thermodynamic properties and how much heat is evaluated by a particular reaction in a reactor. This course introduces chemical engineering thermodynamic theory and applications in the areas of volumetric properties of fluids, heat effects, thermodynamic properties of fluids, thermodynamics of solutions, and physical and chemical equilibria. The data collected for various parameters such as temperature, pressure, humidity, fugacity, enthalpy etc will help to assign stability and reaction operating conditions as and when require.

Laws of thermodynamics for closed and open system, concepts of entropy, entropy changes, reversible and irreversible processes, equilibrium concept, Maxwell's relations, P-V-T behaviour of pure substances, Heat of reaction and effect of temperature on heat of reaction.

Thermodynamic properties of fluids, their calculations using equations of state, partial molar quantities, fugacity, chemical potential, activity coefficients, free energy estimation, Gibb's Duhem Theorem.

Vapour-liquid, vapour–solid and liquid–liquid phase equilibrium for ideal and non ideal systems, Criteria for chemical reaction equilibrium, flow of compressible fluids in pipes and nozzles.

Refrigeration cycle, carnot refrigerator, gas and vapour compression refrigeration, choice of refrigerants, absorption refrigeration, heat pumps, compressors, single stage and multistage, expansion engines, liquification processes

Textbooks:

1) A text book of chemical engineering thermodynamics, **Narayanan K.V**., PHI
2) Introduction to Chemical Engineering Thermodynamics, **Smith J.M**., McGraw Hill
3) Chemical Engineering Thermo dynamics**, Rao Y.V.C**. University press (INDIA) Ltd.
4) Thermodynamics for Chemical Engineers, **Bett K.E., Rowlinson J.S. and Saville G**., MIT Press America
5) Chemical Engineering Thermodynamics, **Dadge B.F**. McGraw Hill Co.

Course Code : **CMP 204**

Course Title : **Technical Analysis**

Course Credits : **1 credit (0-0-2)**

Pre-requisites : **None**

Overlaps with : **None**

Objectives:
Based on the concepts taught, Student shall be able to apply the basic principles to estimate and analyze industrially important materials.

Outcomes:
1. Describe the basic concepts
2. Analyze the given sample.

Program Outcomes for Chemical Engineering at VNIT:
- **a.** An ability to identify, formulates, and solves engineering problems
- **b.** An ability to design a system, component, or process to meet desired needs
- **c.** An ability to design and conduct experiments, as well as to analyze and interpret data
- **d.** An ability to communicate effectively
- **e.** An understanding of professional and ethical responsibility
- **f.** An ability to function on multidisciplinary teams
- **g.** The broad education necessary to understand the impact of engineering solutions in a global and societal context
- **h.** A recognition of the need for, and an ability to engage in lifelong learning
- **i.** A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
CO1	**H**		**H**	**M**			**M**		
CO2	**H**		**H**			**H**			

Course Content:
The course covers the hands-on experience on most of the basic analysis used in the process industries.

CO 1: Describe the basic concepts and

CO 2: Analyze the given sample

1. Synthesis of Biodiesel from Vegetable Oil: To synthesis biodiesel from vegetable oil and determine its physical properties.
2. Sugar analysis using Benedict's reagent: To determine (semi-quantitatively) concentration of reducing sugar in an unknown sample
3. Gravimetric analysis of a phosphorus-containing fertilizer: To determine the percent phosphorous in a sample of fertilizer using Lab-Gravimetric analysis.
4. Estimation of silica content (analysis of ordinary portland cement: To estimate the silica content (analysis of ordinary portland cement (opc))
5. Depression of freezing point: Study of depression of freezing point of solution.
6. Determination of Saponification value of oil: To Determine the saponification value of oil/fat.
7. Study on Sophisticated instruments: GC, UV-Spectrophotometer and FTIR

Textbooks:

1) Puri, Sharma and Pathania, Principles of Physical chemistry, Vishal Publications, Jalandhar, 1998.

Reference Books:

1) Books related to specific experiment as mentioned in the lab manual

Course Code : **CML 231**
Course Title : **Materials Science and Engineering**
Course Credits (L-T-P) : **3 credits (3-0-0)**

Pre-requisites : **None**
Overlaps with : **None**

Course Objectives:
1) To provide basic knowledge and application of different type of materials.
2) To study short-term and long-term mechanical behavior of materials.
3) To understand the science behind the fracture, fatigue and failure of materials.
4) To provide knowledge on phase diagrams in selection of alloy materials with appropriate composition.
5) To understand the heat treatment methods involved in preparation of materials.

Course Outcomes (CO):
At the end of the course, the student Students will gain knowledge on
CO-1: Atomic structure, atomic bonding and basic principles of materials
CO-2: Mechanical properties of metals and deformation of materials
CO-3: Dislocations and strengthening mechanisms of materials
CO-4: Failure of materials-fracture, fatigue and creep
CO-5: Phase diagrams involved in preparation of binary isomorphous and eutectic Systems

Program Outcomes for Chemical Engineering at VNIT:
a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

CO	**a.**	**b.**	**c.**	**d.**	**e.**	**f.**	**g.**	**h.**	**i.**
CO1	□	□				□			
CO2	□	□	□		□				
CO3	□	□	□	□	□	□			□

CO4	□	□	□	□	□	□			□
CO5	□	□	□	□	□	□			□

Course content:

The course deals with the atomic structure, atomic bonding, characterization of materials, basic material properties, testing of materials, fracture, fatigue and deformation of materials under different loads, time and temperature; heat treatment methods and phase diagrams to prepare alloy materials for its application in proper selection of materials for fabrication of various unit operations and unit processes of chemical engineering.

Introduction: Introduction to materials and their principle properties, Atomic bonding, crystal structure and defects, Basic principles in their selection for fabrication and erection of chemical plant.

Deformation: Plastic deformation - Mechanism of plastic deformation, slip, work hardening, deformation in polycrystalline materials, Effect of cold working and annealing, hot working. Elastic deformation, Anelastic deformation, Viscoelastic deformation – models for viscoelastic behavior.

Fracture: Types of fracture, cleavage, brittle, ductile, Griffith crack theory, Theories of crack initiation, ductile –brittle transition.

Testing of materials: Destructive tests - Tensile testing, stress-strain curves, condition for necking, compression testing, Hardness testing. Creep - testing method, creep curve, requirements for creep resistance materials. Fatigue – testing method, fatigue prevention. Non destructive tests.

Thermal properties: Heat capacity and specific heat, Thermal expansion, thermal conductivity, thermal shock.

Heat treatment: Annealing, quenching, normalizing, hardening, martempering, austempering, case hardening, cyaniding, nitriding, flame hardening, induction hardening, diffusion coating, furnaces and temperatures.

Phase diagram: Basic terms, Hume - Rothery rules of solid solubility, Gibb’s phase rule, polymorphism, solidification of pure metal. Types of cooling curves, plotting of equilibrium diagram, lever rule, common types of phase diagram, other transformations in alloy system; Non-equilibrium cooling.

Engineering materials – overview: Organic, inorganic and biological materials: Polymers, Metals & alloys, ceramics, glasses, leather, wood. Electronic materials, photonic materials, smart materials, composites and nanostructured materials.

Textbooks:

1) Material Science and Engineering-An Introduction (8th edition), William D. Callister, Jr. John -Wiley and sons, USA
2) Introduction to material science, James F. Shacketford, McMillan publishing company, New York ISBN 1990.
3) William D. Callister Jr., “Material Science and Engineering”, John Wiley and Sons, Inc.,
4) Properties of Engg. Materials, Jestrazebaski D.Z., Toppers. Co. Ltd. 3rd edition.
5) Askeland, Fulay, Wright and Balani “The science and Engineering of materials”, Cngage learning publishers, 6th edition, 2012, ISBN 978-81-315-1641-6
6) Material science and processes, Hajra Choudhury S. K and Hajra Choudhury A. K, Indian book distributing corporation.

Course Code : CML 221
Course Title : Mass Transfer I
Course Credits (L-T-P) : 4 (3-1-0)
Pre-requisite : CML 262
Overlap with **:** NIL

Course outcome/objectives:

1. To understand the concept of diffusion for gas, liquid and solid phase.
2. To understand the concept of interphase mass transfer.
3. To design the equipments for mass transfer operations.
4. To understand the processes and design the equipments for drying, adsorption, crystallization, and humidification.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference)

PO → / CO ↓	A	b	c	d	e	f	g	h	i	j	k
CO1	H	H	M	-	-	-	L	-	-	-	-
CO2	H	H	M	-	-	-	L	-	-	-	-
CO3	H	H	H	-	-	-	L	-	-	-	-
CO4	H	H	H	-	-	-	L	-	-	-	-

Content (CO wise):

CO1: Introduction to mass transfer operations, Diffusion in gases, liquids, and solids. Steady state and unsteady state operations. Individual and overall mass transfer coefficients. CO2: Theories, analogies of mass transfer. Inter-phase mass transfer operation;

CO3: Design of gas, liquid and solid contact equipments like distillation column, spargers, scrubbers, wetted wall tower, fractionation tower and their types.

CO4: Drying: Constant rate and falling rate periods, equilibrium moisture contents, drying equipments, rotary dryers, drum dryers, vacuum dryers, Spray dryer, fluidized bed dryers, dryer calculations and dryer selection criteria.

Crystallization: Theory of Crystallization, saturation, supersaturation, nucleation and crystal growth, various equipments for crystallization, their operational and design characteristics.

Adsorption: Adsorption isotherms, adsorption agents, equipments for adsorption, pressure swing adsorption technology, adsorption phenomena

Humidification: Equipment's operational characteristics, design procedures and selection criteria along with mass transfer calculations, Types of cooling towers, cooling tower operational characteristics.

Text Books/ Reference Books:

1) Mass Transfer Operations, Treybal R.E., McGraw Hill Book Co., New York 1980, 3rd Edition
2) Chemical Engineering Vol. I, II & III, Coulson J.M. and Richardson J.F., Pergamon Press, New York 1977
3) Unit Operations of Chemical Engineering, McCabe W.L. and Smith J.C. & Harriot, McGraw Hill Book Co., New York 1980, 5th Edition
4) Principles of Mass Transfer and Separation Process, Binay K. Dutta, PHI Learning Pvt. Ltd., New Delhi, Eastern Economy Edition
5) Introduction to Chemical Engineering, Badger W.L. and Banchero J.T., Tata McGraw Hill Book Co.
6) Transport processes and separation process principles, C. J. Geankoplis, PHI publication

Course Code : CML222
Course Title : Heat Transfer
Course Credits (L-T-P) : 4 (3-1-0)
Pre-requisites **:** Maths, CML204-(CET), CML202-Fluid Mechanics
Overlaps with **:** None

Objectives: To study different modes of Heat Transfer and its application in various chemical industries

Course Outcomes

1) The objective of this course is to introduce the basic principles of heat transmission by conduction, convection, and radiation
2) To identify, formulate, and solve engineering problems involving conduction, convection, and radiation,
3) To apply energy balances and rate equations to model and analyze thermal systems.
4) To introduce application of heat transfer and its optimum use

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with Pos:

POs → / Cos ↓	a	b	c	d	e	f	g	h	i	j	k	l
CO1	H	L	L	L	L	H	M	L	M	M	L	L
CO2	H	H	M	L	L	M	L	L	L	M	L	L
CO3	H	H	H	L	L	M	L	L	M	L	M	M
CO4	M	L	M	L	L	H	M	L	M	L	M	M
Overall	H	H	M	L	L	M	M	L	M	M	M	M

Description of the content (CO wise):

CO.1. Basic modes of heat transfer, Fourier's law, thermal conductivity, steady state heat conduction through a plane, composite wall, cylinder, sphere, heat generation inside solids, unsteady state heat conduction, types of thermal insulation, critical thickness and optimum thickness of insulation, extended surfaces, fin performance evaluation, effectiveness of fins.

CO.2. Free and forced convection inside and outside the tubes as well as over the plates, individual and overall heat transfer coefficients. Heat transfer in laminar flow and turbulent flow, dimensional analysis, dimensionless numbers in heat transfer, heat transfer correlations for natural convection.

CO.3. Condensation and Boiling, Condensation over flat plate, condensation inside and outside the tubes in horizontal, vertical and inclined position, film condensation, dropwise condensing. Estimation of film coefficient of heat transfer for condensing vapours turbulence in condensing film. Heat Transfer to boiling liquids, pool boiling and forced convection boiling, boiling curve and its characteristics.

CO.4. Radiation heat transfer, laws of radiation, concepts of black body, gray body, green house effect, emissive power, heat flux by radiation, view factors, radiation shield, luminous and non luminous gases, heat exchangers, heat transfer fluids.

Textbooks/ Reference Books:

1) Hollman J.P.; Heat Transfer, McGraw Hill, 1993
2) Incropera F.P. Fundamentals of Heat and Mass Transfer 5th Edition Wiley India Pvt.Ltd Ltd.,2008
3) Cengel Y.A. Heat Transfer: A Practical Approach McGraw-Hill; 2 edition ,2002
4) Kern D.Q., Process Heat Transfer, Tata McGraw Hill Book Co., New Delhi, 1990.
5) Coulson J.M., Richardson J.R. Chemical Engineering, Vol. I 5th Edition, Butterworth Heinemann, New Delhi.
6) Dutta B.K. Heat Transfer; Principles and Applications PHI Pvt.Ltd New Delhi ,2006

Course Code : CML 223
Course Title : Chemical Reaction Engineering-I
Course Credits (L-T-P) : 4 (3-1-0)
Pre-requisites : None
Overlaps with : None

Objectives:

At the end of the course, the student will understand the reaction kinetics, reaction mechanism and reaction path of a given chemical reaction, type and sequential arrangement of the reactors to be used.

Course Outcomes:

1) Provide students with a basic understanding of reaction engineering, type of reactions, reaction kinetics.
2) Enhance their knowledge on types of reactors, working of reactors and different types of arrangements of reactors
3) Improve the student's ability in deciding type of reactor and their arrangement for achieving desired conversion keeping economy in point of view

Relationship of Course Objectives to Program Outcomes:

Course objectives	Program outcomes (POs)								
	a	b	c	d	e	f	g	h	i
CO1	✓					✓		✓	
CO2	✓					✓		✓	
CO3	✓	✓				✓		✓	

Course description:

CO.1. Introduction to Chemical Reaction Engineering: Basics of chemical reaction engineering, Role of Chemical Reaction Engineering in Process Industry, Classification of reaction based on various terms, Reaction rate, Chemical kinetics, Variables affecting rate of reaction, Speed of reactions, Problems.

CO.2. Kinetics of Homogeneous Reactions: Concentration dependent term and temperature dependent terms of rate equation, Single and multiple reactions, Elementary and non-elementary reactions, Molecularity and order of reaction, Rate constant, Representation of reaction rate, Kinetic models, Temperature dependency from Arrhenius' law, thermodynamics, various theories, Activation energy, Problems.

CO.3. Interpretation of Batch Reactor Data: Constant volume batch reactor, Variable volume batch reactor, Integral method and differential method of analysis of kinetic data, other methods of analysis of kinetic data, Temperature and reaction rate, Problems.

Course Content:

1. Introduction To Reactor Design: Types of reactors, PFR, CSTR etc., Material & energy balances single ideal reactor, Space-time and space-velocity, Holding time, Introduction of non-ideal flow, Problems
2. Ideal Reactors for a Single Reaction: Ideal Batch Reactor, Steady State Mixed Flow Reactor, Steady State Plug Flow Reactor, Problems
3. Design for Single Reactions: Size comparison of single reactors, General graphical comparison, Multiple reactor system, Recycle reactor, Autocatalytic reactions, Problems.
4. Design for Parallel Reactions: Introduction to design of parallel reactions, Qualitative and Quantitative discussion on product distribution, Contacting patterns, Reactor Size and arrangement, Selectivity, Yield, Problems.
5. Potpourri of Multiple Reactions: Reversible first order reaction, First order followed by zero order reaction, Zero order followed by first order reaction, Successive reversible reactions of different orders, reversible reactions, Irreversible series-parallel reactions, Graphical representation, Denbigh reactions and their special cases, Problems.
6. Temperature and Pressure Effects: Single and multiple reactions, Heats of reaction from thermodynamics, Equilibrium constant, Temperature, Graphical design procedure, Optimum Temperature Progression, Heat Effects, Adiabatic and non-adiabatic operations, Problems.

Textbooks:

1. Chemical Reaction Engineering, Octave Levenspiel, John Wiley & Sons, Singapore, 3rd edition, 1998.
2. Elements of Chemical Reaction Engineering, Fogler H.S., Prentice-Hall, NJ, 4th edition, 2006.
3. Chemical Reactor Analysis, G. F. Froment and K. B. Bischoff, John Wiley & Sons, Singapore, 2nd edition, 1990.
4. Chemical Engineering Kinetics, Smith J. M., McGraw Hill, N Y, 3rd edition, 1981.

Course Code : CML224
Course Title : Chemical Engineering Mathematics
Course Credits (L-T-P) : 3 (3-0-0)
Pre-requisites : None
Overlaps with : None

Objectives: To study different optimization technique

Course Outcomes

1) Able to solve linear algebraic equations.
2) Able to solve non-linear algebraic equations.
3) Able to solve ordinary differential equations.
4) Able to formulate the problem and apply discretisation techniques.
5) Able to formulate and solve optimization problems.

Mapping with POs (Departmental reference) *:

POs → / COs ↓	A	b	c	d	E	F	g	h	i	j	k
CO1	H	H			H				H		
CO2	H	M			H				M		
CO3	H	M	M	M	H	L			H		
CO4	M	H	M	M	M	M			M	M	
CO5	H	H	H	H	H	M			H	M	
Overall	H	H	M	M	H	M			H	M	

Content (CO wise):

CO.1. *Linear Algebraic Equations:* System of linear algebraic equations, Classification of solution approaches as direct and iterative: Gaussian elimination method, Introduction to methods for solving sparse linear systems: Thomas algorithm or tridiagonal matrix algorithm, Iterative methods: Derivation of Jacobi, Gauss-Siedel and successive over relaxation methods, System of linear equation using eigen values and eigen vector. Examples using Comsol Multiphysics Software on Sequence of distillation columns, Reactors linked with pipes, Rectangular fin, Recovery of acetone, etc.

CO.2. *Nonlinear Algebraic Equations:* Method of successive substitutions derivative free iterative solution approaches, Secant method, regula falsi method and Wegsteine iterations, Modified Newton's method and qausi Newton method with Broyden's update, Optimization based formulations and Leverberg Marquardt method, Examples using Comsol Multiphysics Software on Pressure Drop in Pipe, Minimum Fluidization Velocity, Terminal Velocity, Solution of Cubic Equations of State, Bubble Point and Dew Point Temperature Calculations Using Raoult's Law, Bubble Point and Dew Point Temperature Calculation, etc.

CO.3. *Ordinary Differential Equations – Initial Value Problems (ODE-IVPs):* Analytical Solutions of Linear ODE-IVPs, Basic concepts in numerical solutions of ODE-IVP: step size and marching, concept of implicit and explicit methods, Taylor series based and Runge-Kutta methods: derivation and examples, Multi-step (predictor-corrector) approaches: derivations and examples, Introduction to solution methods for differential algebraic equations (DAEs), Single shooting method for solving ODE-BVPs, multiple ordinary differential equation, Examples using Comsol Multiphysics Software on Double Pipe Heat Exchanger, Stirred Tank with Coil Heater, Pneumatic Conveying, Series of Stirred Tanks with Coil Heater, Batch and Stirred Tank Reactors, Plug Flow Reactor, Nonisothermal Plug Flow Reactor, etc.

CO.4. *Problem Discretization Using Approximation Theory:* Finite difference method for solving ODE-BVPs , Finite difference method for solving PDEs, Model Parameter Estimation using linear least squares method, Gauss Newton Method, Curve fitting and regression, Examples using Comsol Multiphysics Software on One-Dimensional Steady Heat Conduction, Chemical Reaction and Diffusion in Pore, etc.

CO.5. *Optimization:* Basic concept of optimization and formulation, Nature of optimization problem (constraint and unconstraint), Linear programming by simplex method, Application of optimization based on simplex method.

Textbooks:

1. Chapra S C, Applied Numerical Methods with Matlab for Engineers and Scientists, McGraw Hill, 1st Edition, 2005.
2. Gupta S K, Numerical Methods for Engineers. Wiley Eastern, New Delhi, 1st Edition, 1995.
3. Ahuja P, Introduction to Numerical Methods in Chemical Engineering. Prentice Hall India, New Delhi, 1st Edition, 2010.

Reference Books:

1. Ghoshdastidar P S, Computer simulation of flow and heat transfer, Tata McGraw-Hill Publishing, 1st Edition, 1998.
2. Pushpavanam S, Mathematical Methods in Chemical Engineering. Prentice Hall India, New Delhi, 4th Edition, 2009.
3. Thomas F and Himmelblau D M, Optimization of Chemical Processes, McGraw Hill, 2nd Edition, 2001.
4. Finlayson B A, Introduction to Chemical Engineering Computing, Wiley, 2nd Edition, 2014.

Course Code : CML 225
Course Title : Chemical Engineering Thermodynamics - II
Course Type : Core
Course Credits : 4 (4 -0-0)
Pre-requisites : CML 204: Chemical Engineering Thermodynamics - I
Overlaps with : None

Objectives:
Based on the concepts taught, Student shall be able to apply the concepts of thermodynamics to solve the given problem on thermodynamic process accurately.

Outcomes:
1) Describe the Thermodynamic properties like free energy, activity, fugacity etc., for pure substances and solve the related problems.
2) Describe the thermodynamic properties of solutions and solve the related problems.
3) Use the concepts of phase equilibrium and solve the problems.
4) Apply the concepts of chemical equilibrium and solve the problems.

Program Outcomes for Chemical Engineering at VNIT:
- **a.** An ability to identify, formulates, and solves engineering problems
- **b.** An ability to design a system, component, or process to meet desired needs
- **c.** An ability to design and conduct experiments, as well as to analyze and interpret data
- **d.** An ability to communicate effectively
- **e.** An understanding of professional and ethical responsibility
- **f.** An ability to function on multidisciplinary teams
- **g.** The broad education necessary to understand the impact of engineering solutions in a
- **h.** global and societal context
- **i.** A recognition of the need for, and an ability to engage in lifelong learning
- **j.** A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	H			H					
2	H	M					H		
3	H	M					H		
4	H	M					H		

Course Content: The course present thermodynamics from a chemical engineering viewpoint. Thermodynamic properties, Concepts of phase and chemical equilibria and its application are stressed in this course.

CO 1: Describe the Thermodynamic properties like free energy, activity, fugacity etc., for pure substances and solve the related problems.

Thermodynamic properties of pure substances: fugacity, fugacity coefficient, compressibility factor, activity.

CO 2: Describe the thermodynamic properties of solutions and solve the related problems. Solution thermodynamics: Ideal and non-ideal gas mixtures and liquid solutions, partial molar properties, physical significance and determination methods, chemical potential.

Gibbs-Duhem equation: General form, various forms of Gibbs-Duhem equation, applications, limitations; Property changes of mixing, excess properties.

CO 3: Use the concepts of phase equilibrium and solve the problems. Criteria of phase equilibrium, Duhem theorem. Vapour liquid equilibrium, VLE equation, low pressure VLE, Phase diagrams for binary solution, T-x-y and P-x-y diagrams. Effect of pressure on VLE. Azeotropes and its types. Activity coefficient; equations used for the determination, Margules, van Laar, Wilson equations, VLE at high pressures, bubble point, dew point calculations, Thermodynamic consistency tests for VLE data.

CO 4: Apply the concepts of chemical equilibrium and solve the problems.

Chemical reaction equilibrium; criteria of equilibrium, Reaction stoichiometry, equilibrium constant, Gibbs free energy change, choice of standard state, feasibility of chemical reactions, effect of temperature on equilibrium constant, evaluation of van't Hoff constant, Effect of parameters like temperature, pressure, composition on the equilibrium conversion.

Textbooks:

1) Smith J.M., Van Ness H. C., and Abbott M.M., Introduction to Chemical Engineering Thermodynamics, McGraw Hill, 2005.
2) Narayan K.V., a text book of Chemical Engineering Thermodynamics, PHI, 2001.

Reference Books:

1) S.Sandler, Chemical, biochemical and engineering thermodynamics, 4th edition, John Wiley, 2006.
2) Rao Y.V.C., Chemical Engineering Thermo dynamics, University press (INDIA) Ltd.
3) Hill, T.L., An Introduction to Statistical Thermodynamics, Dover Publications,1960

Course Code	: CMP202
Course Title	: Fluid Mechanics
Course Credits	: 1(0-0-2)
Pre-requisites	: None
Overlapping/Equivalent courses	: None

Course Objectives:

1) To impart the basic concepts of fluid mechanic

2) To develop understanding about flow measurement device

3) Understand frictional losses in pipes

4) The students are exposed to basic theory, calculations and machinery involved in various fluid handling operations

Course Outcomes:

1) To understand importance of various Fluid mechanics concept used in process industry.

2) To apply principles of basic sciences and chemical engineering for designing various Flow and pressure measuring instruments

3) To experience of handling different unit operations

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems

b. An ability to design a system, component, or process to meet desired needs

c. An ability to design and conduct experiments, as well as to analyze and interpret data

d. An ability to communicate effectively

e. An understanding of professional and ethical responsibility

f. An ability to function on multidisciplinary teams

g. The broad education necessary to understand the impact of engineering solutions in a global and societal context

h. A recognition of the need for, and an ability to engage in lifelong learning

i. A knowledge of contemporary issues

Mapping POs (Departmental references)

Course Outcomes	**Program outcomes**								
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
1	**H**			**H**		**M**	**H**	**M**	
2	**H**		**H**	**H**			**H**	**M**	
3	**H**		**H**	**H**	**L**	**M**	**H**	**H**	**M**

Course description:

The course covers the Hands-on experience of working by conducting experiments on most of the basic unit operations like venturi-meter, orificemeter, notches, friction in pipes and measurement of pressure through manometers

Course Content (CO wise)

Expt. No.	Details
1	**Friction in Pipeline** To determine the friction coefficient factor for different size as well as different material of pipes.
2	**Flow Through Notches** To determine discharge coefficient C_d through different Notches.
3	**Packed Bed** Hydraulic study in Packed column.
4	**Fluidized Bed** To study the relation between the pressure drop & superficial velocity throughout the bed.
5	**Pressure Drop In Fittings** To study the losses in pressure due to different fitting in pipes.
6	**Impact of Jet on Vanes** To study flow of fluid through Jet.
7	**Pressure Measurement** Pressure measurement in pipeline.
8	**Two Phase Flow** To study the flow pattern of two phase in singe tube.
9	**Orifice Meter** To determine discharge coefficient C_d.
10	**Venturi Meter** To determine discharge coefficient C_d.
11	**Pitot Tube** To determine discharge coefficient C_d & determine point velocity.

Text Books:

1) Chattopadhyay O.P., Unit Operations of Chemical Engineering, Vol. 1 & 2, Khanna Publications, New Delhi, 1996.

2) Badger & Banchero Introduction to Chemical Enginerring Mc-Graw- Hill Education

3) G.G. Brown Unit Operation John Willey

4) Hiremath R.S & Kulkarni A.P.,. Mechanical Operations Vol I Everest Publication

Reference Books:

1) McCabe W.L., Smith J.C. and Harriot P., Unit Operations of Chemical Engineering McGraw Hill, New York 2001. 6th Edition

2) Coulson J. M. and Richardson J.F; Chemical Engineering Vol. 1& 2 Publishers: Butter worth – Heinemann Ltd. 2001-2002.

3) Christie J. GeanKoplis Transport processes & Unit Operation Prentice Hall International

Course Code : **CMP203**
Course Title : **Mechanical Operation**
Course Credits : **1(0-0-2)**
Pre-requisites : **None**
Overlaps with : **None**

Course description:

The course covers the Hands-on experience of working by conducting experiments on most of the basic unit operations like hydraulic classifier, sedimentation, ball mill, jaw crusher, cyclone separator, filtration equipment and sieve analysis.

Objectives:

1. To impart the basic concepts of mechanical operations
2. To develop understanding about size analysis, size reduction and solid handling
3. Understand mechanical separation methods such as filtration, sedimentation, transportation of solids etc and associated equipments used for achieving these methods
4. The students are exposed to basic theory, calculations and machinery involved in various solid handling operations

Outcomes:

1. To understand importance of various mechanical operations used in process industry.
2. To apply principles of basic sciences and chemical engineering for designing various size reduction, size separation and conveying equipments.
3. To experience of handling different unit operations

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	**H**			**H**		**M**	**H**	**M**	
2	**H**		**H**	**H**			**H**	**M**	
3	**H**		**H**	**H**	**L**	**M**	**H**	**H**	**M**

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Text Books:

1) Chattopadhyay O.P., Unit Operations of Chemical Engineering, Vol. 1 & 2, Khanna Publications, New Delhi, 1996.
2) Badger & Banchero Introduction to Chemical Enginerring Mc-Graw- Hill Education
3) G.G. Brown Unit Operation John Willey
4) Hiremath R.S & Kulkarni A.P.,. Mechanical Operations Vol I Everest Publication

Reference Books:

1) McCabe W.L., Smith J.C. and Harriot P., Unit Operations of Chemical Engineering McGraw Hill, New York 2001. 6th Edition
2) Coulson J. M. and Richardson J.F; Chemical Engineering Vol. 1& 2 Publishers: Butter worth – Heinemann Ltd. 2001-2002.
3) Christie J. GeanKoplis Transport processes & Unit Operation Prentice hall International

Course Code : CMP 322
Title of the Course : Heat Transfer
Course Credits (L-T-P) : 1 Credits (0-0-2)
Pre-requisites: CML 222
Overlapping/Equivalent courses : None

Course Objectives/Outcomes:

1) To give the in-hand experience of lab scale experiments on various equipments such as heat transfer through forced convection, pin fin, lagged pipe, emissivity apparatus, stefan'sboltzmann apparatus, shell and tube heat exchanger, double pipe heat exchanger, open pan evaporator, single effect evaporator, heat transfer in agitated vessel system
2) To observe and note down the steady state temperatures of all equipments.
3) To determine the heat transfer rate, heat transfer coefficient, and overall heat transfer coefficient for various equipments such as shell and tube heat exchanger, double pipe heat exchanger etc.
4) To introduce application of heat transfer equipments and its optimum use

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs→ / Cos↓	a	b	c	d	e	f	g	h	i	j	k	l
CO1	H	L	L	L	L	H	M	L	M	M	L	L
CO2	M	H	M	L	L	M	L	L	L	M	L	L
CO3	H	H	H	L	L	M	L	L	M	L	M	L
CO4	M	L	M	L	L	H	M	L	M	L	M	L
Overall	H	H	M	L	L	M	M	L	M	M	M	L

Course content (CO wise):

CO.1. To find surface heat transfer coefficient for a pipe flowing heat by forced Convection of air flowing through it for different air flow rate and heat flow rate
CO.2. To study the temperature distribution along the length of a pin fin under free and forced convection heat transfer
CO.3. To determine heat flow rates through the lagged pipe for known value of thermal conductivity of lagged material and To plot the temperature distribution across the lagged material.
CO.4. To determine the emissivity of grey surface.
CO.5. To find out Stefan's Boltzmann constant.
CO.6. To determine cold water side and hot water side heat transfer coefficient, LMTD and overall heat transfer coefficient for parallel and counter flow.
CO.7. To calculate rate of heat transfer, LMTD and overall heat transfer coefficient for parallel and counter flow.
CO.8. To determine the evaporation coefficient and overall heat transfer coefficient of the open pan.
CO.9. To determine the overall heat transfer coefficient of the evaporator.
CO.10. To determine the heat transfer coefficient in agitated vessel system.

Textbooks/ Reference Books:

1) Coulson J.M., Richardson J.R. Chemical Engineering, Vol. I, Butterworth Heinemann, New Delhi.
2) Kern D.Q, Process Heat Transfer, Tata McGraw Hill Book Co., New Delhi, 1990.

Course No. : CML233

Course Title : Environmental Engineering
Course Type : Elective
Course Credits (L-P-T) : 3(3-0-0)
Pre-requisites : None
Overlaps with : None

Brief Description: Environmental engineering will cover the types of pollutants, nutrient cycle, effect of pollutants on man and environment and the methods for prevention and treatment of pollutants.

Course Objectives:
1. To identify the sources of pollutants and its hazards on human and environment
2. To learn the nutrient cycle, ecosystem and ways to restore the ecosystem and cycles
3. To learn the methods for prevention and control / treatment of various pollutants
4. To understand the life cycle and the sustainability concept related to environment
5. To learn the basic environmental legislation

Course Outcomes:
CO1: Understand key current environmental problems.
CO2: Be able to analyse an industrial activity and identify the environmental problems.
CO3: Be able to plan strategies to control, reduce and monitor pollution.
CO4: Be able to select the most appropriate technique to purify and/or control the emission of pollutants.
CO5: Be conversant with basic environmental legislation.

CO	PO											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2			3	3			2	3
CO 2	3	3	3	3			3	3				
CO 3	3	3	3	3	1	2	3	3				3
CO 4	3	3	3	3	1	2	3	3	1			
CO 5				3			3	3			3	3

Expanded Course Description:
Man and environment, biogeochemical cycles, Biosphere and ecosystem, Forest Nutrient cycles and the parameters responsible for the disturbance of these cycles.

Mobile and stationary sources of air pollutants, air pollution, behaviour of pollutants and atmospheric chemical reactions, Atmospheric dispersion of pollutants, models for dispersion, limitations of models, effective stack height concept, gas sampling and analysis. CO, CO_2, H_2S,

SO_x, NO_x emissions and their control, desulphurization. Sources of water and pollutants, classification and characterization of solid, liquid and gaseous waste.

Air pollution control processes, Unit operations such as screening, coagulation, flocculation, filtration, clarification, solvent extraction. measurement of levels of pollution such as DO, BOD, COD, TOC,ThOD, soluble and suspended volatile solids, Water quality and discharge standards

Chemical treatment of waste material, oxidation, chlorination, Ozonation, incineration etc. Biological treatment, biochemical kinetics, microbial kinetics, microbial growth. Aerobic and anaerobic waste treatment, activated sludge process, aerated lagoons, anaerobic digesters, Biogas & trickling filters & its utilization. Solid Waste & its disposal pyrolysis (Incineration, Composing and filling etc.).

Measuring environmental impacts, life cycle analysis, legislation controlling discharges, optimal degree of abatement, and policies for regulation of environmental impacts.

Text Books:

1) Mahajan S.P. Pollution Control in Process Industries, Tata McGraw Hill Book Co.
2) Pandey G.N. and Camey G.C.; Environmental Engineering, Tata – McGraw Hill Book Co., New Delhi (1989)
3) Rao C.S.; Environmental Pollution Control Engineering, New Age International Publishers, 3rd Edition
4) Sincero A., Environmental Engineering, A Design Approach. , Prentice – Hall of India, New Delhi (1996).

Course Evaluation:

- Class Tests and Assignments: 20%
- Mid-Semester Exam (1.5 Hours): 30%
- End-Semester Exam (3 Hours): 50%

Course No. : CML234
Course Title : Industrial Waste Treatment
Course Credits (L-T-P) : 03 credits (3-0-0)
Pre-requisites **:** None
Overlaps with : None

Objectives:

Our objective is to discuss engineering aspects of industrial Pollution Control Technologies so that student shall be able to apply different control strategies to protect the ecosystem.

Outcomes:

1) Student will gain the knowledge of different pollutants and their effects on public health and on ecosystem.
2) Students will learn the different methods of sampling and measurement of air, water and solid pollutants.
3) Students will gain the knowledge of conventional methods as well as new methods to control the pollution.
4) Students will able to identify air, water, and solid waste management practices and technologies adapted by different industries.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. Recognition of the need for, and an ability to engage in lifelong learning.
i. A knowledge of contemporary issues

Course Objectives	**Program Outcomes**								
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
1			**M**			**H**		**L**	
2		**H**							**L**
3	**H**	**M**						**L**	
4	**H**	**M**				**M**		**L**	
Overall	**H**	**L**	**M**			**L**		**L**	**L**

Course description:

Our objective is to discuss engineering aspects of industrial Pollution Control Technologies. The emphasis in this course will be the control of gaseous , liquid and solid pollutants.

Course content:

Nature and characteristics of industrial wastes; Sources and types of wastes: solid, liquid, and gaseous wastes; Pre-treatment of Industrial wastes, unit operations and unit processes. Sampling Techniques.

Methods for Treating industrial waste gases or air discharges- physical method, chemical method, combined method, biological method.

Solid and Hazardous wastes: definitions, concepts. Incineration, recycling, composting, landfill, On-Site Monitoring and Analysis of Industrial Pollutants.

Waste water treatment-physical, chemical and biological method.

Recent trends in Industrial waste treatment. Application of Biotechnology for Industrial Waste Treatment.

Case Studies:- Example (Treatment of Pharmaceutical Wastes, Treatment Refinery Wastes, Treatment of Textile Wastes, Treatment of Pulp and Paper Mill Wastes, Treatment of Dairy Processing Wastewaters, Treatment of Pesticide Industry Wastes, Food Waste Treatment, Treatment of Rubber Industry Wastes, Treatment of Tannery Industry Wastes and Radioactive waste etc.).

Textbooks:

1) Pollution Control in Process Industries,S. P. Mahajan,Tata McGraw Hill ,1st
2) Environmntal pollution control engineering,C.S.Rao,New Age International, 2nd

Reference Books:

1) Industrial Pollution Prevention Handbook, Freeman H. M.,McGraw Hill,1st
2) Industrial Waste Treatment Handbook,FrankWoodard,Butterworth-Heinemann, 1st

Industrial Waste Treatment,Nemerow N.L,Butterworth-Heinemann,1st

Course Code **:** CML 392
Course Title **:** Corrosion Engineering
Course Credits (L-T-P) **:** 3 (3-0-0)
Pre-requisites **:** None
Overlap with **:** None

Objectives:

At the end of the course, the student will understand the reaction kinetics, types and thermodynamics of corrosion, its preventive methods and may able to devise novel equipments from MOC point of view to prevent corrosion.

Course Outcomes:

1. Provide students with a basic understanding of corrosion phenomena, its types, reaction kinetics and thermodynamics.
2. Enhance their knowledge on industrially important corrosion problems, factors responsible for corrosion, its preventive methods and minimizing losses due to corrosion
3. Improve student's ability in handling on site industrial corrosion problems and devise novel equipments from MOC point of view.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Relationship of Course Objectives to Program Outcomes:

Course objectives	Program outcomes								
	a	b	c	d	e	f	g	h	I
CO1	✓					✓		✓	
CO2	✓					✓		✓	
CO3	✓	✓				✓		✓	

Course description:

1. Thermodynamics and Kinetics of electrochemical corrosion: Theory of corrosion, reaction mechanisms and reaction kinetics, thermodynamic aspects of corrosion, Nernst equation. Basic

wet corrosion cell, electrode potential, potential-ph diagram, Butler-Volmer equation, polarization, mixed potential theory, passivity, immunity.
2. Types of corrosion – recognition and mechanisms: Uniform corrosion, galvanic corrosion, pitting, dealloying, crevice corrosion, intergranular corrosion, stray current corrosion, high temperature corrosion, flow-assisted corrosion, cavitations corrosion, fretting corrosion cracking process, Microbially influenced corrosion.
3. Corrosion measurements: Methods of measurement s of corrosion based on study of various ASTM standards for corrosion–Weight, electrochemical, electrical, thickness etc.
4. Corrosion protection and surface engineering: Principles of different methods of corrosion protection and surface treatment, chemical and electrochemical surface treatments if metals. Protective coatings like plating, pvd, cvd, thermal spray, hot dip, applications of inhibitors, and electrochemical methods for corrosion protection.
5. Other Corrosion Environment and Materials Selection: selection of corrosion resistant materials for use in acids, alkalies, atmosphere, soils, seawater, freshwater, etc.

Textbooks:

1) Butlar G. and Ison' HC. K ",'Corrosion and its Prevention in Waters, Leonard Hill- London (1966).
2) Rajgopalan, K.S. ", Corrosion and its Prevention, Chemical Engineering Education, Development Centre, I.IT. Madras (1975).

Uhlig H. H, Corrosion and Corrosion control, John Wiley and sons (1971).

Course Code	: CML 301
Course Title	: Mass Transfer - II
Course Credits (L-T-P)	: 3 (3 -0-0)
Pre-requisites	: None
Overlaps with	: None

Objectives:

The objective of this course is to introduce the mass transfer operations and how to quantify, formulate, and solve engineering problems involving different mass transfer operations like distillation, leaching, liquid-liquid extraction, absorption. To demonstrate that how to apply mass balances and its transfer and analyze systems.

Course Outcomes:

1) To understand the basic concepts and principles of mass transfer operation like distillation, absorption, liquid-liquid extraction, solid-liquid extraction (leaching).
2) How to identify, quantify or formulate, and solve the engineering problems involving mass transfer.

3) Design and analysis of mass transfer processes and equipments.
4) Applications of mass transfer phenomena and its optimum use.

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
CO1	H	M	L	H		M			
CO2	H	M	H			L	M		L
CO3	H	M		L	M	L			
CO4	H	M	L	M	L	L	M	L	L

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Course Content: The course presents the mass transfer operations from a chemical engineering viewpoint. The basic concepts, design and its application regarding the mass transfer operations like Distillation, Absorption, Liquid-liquid extraction and Solid-liquid extraction are stressed in this course.

Unit I (6 hrs)

Distillation: Vapour – liquid equilibria, Raoult’s law, X-Y and H-X-Y diagrams, differential distillation and equilibrium distillation, steam distillation, azeotropic distillation, extractive distillation.

Unit II (8 hrs)

Fractionation, binary distillation, plate and packed columns for distillation, analytical and graphical methods for estimation of number of stages required in distillation column, minimum reflux ratio, optimum reflux ratio, number of stages at optimum reflux, murphree plate efficiency and overall plate efficiency, effect of feed conditions on number of plates for separation.

Unit III (4 hrs)

Concept of HETP, HTU, NTU in distillation, plate and packed columns, packings for packed columns, pressure drop in plate and packed columns, bubble cap, sieve tray, valve tray plate columns.

Unit IV (6 hrs)

Absorption Equilibrium relationships, two film theory, penetration theory, surface renewal rate theory, concept of driving force and mass transfer coefficient, plate column and packed columns for absorption, selection of solvent for absorption and absorbers design procedures.

Unit V (6 hrs)

Liquid – Liquid Extraction fundamentals, selection of solvent for extraction, estimation of mass transfer coefficients, triangular diagram representation, equipments for liquid – liquid extraction, plate and packed columns, spray columns, rotary disc contactors, design procedures and equipment selection criteria. Single stage, multistage operations etc.

Unit VI (6 hrs)

Solid – Liquid Extraction fundamentals, Solvent selection, equilibrium relationship, triangular diagram representation, single stage, multistage concurrent and counter current operation, equipments for solid – liquid extraction, their design procedure and selection criteria.

Textbooks:

1) Treybal R.E., Mass Transfer Operations, McGraw Hill Book Co., New York 1980.
2) Coulson J.M. and Richardson J.F., Chemical Engineering Vol. I, II & III, Pergamon Press, New York 1977.

Reference Books:

1) Badger W.L. and Banchero J.T., Introduction to Chemical Engineering, 4th edition, Tata McGraw Hill Book Co..
2) McCabe W.L. and Smith J.C. & Harriot, Unit Operations of Chemical Engineering, 5th edition, McGraw Hill Book Co., New York 1980.
3) Binay K. Dutta, Principles of Mass Transfer and Separation Process, PHI Learning Pvt. Ltd., New Delhi.

Course Code : CML 302
Title of the Course : Chemical Process Modelling and Simulation
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : CML 263, CML361, CML366, CML362, CML367
Overlapping/Equivalent courses : None

Course Objectives/Outcomes:

1) To understand knowledge of fundamental principles and basic laws of modeling
2) To understand the approach for mass/heat transfer & CRE
3) To apply the knowledge of differential equations
4) To understand the approach for modeling
5) Formulation of mathematical model for various chemical Engg. system

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs → / COs ↓	a	B	c	d	e	f	g	h	i
CO1	√					√	√		
CO2	√					√	√		
CO3	√					√	√		
CO4	√					√	√		
CO5	√					√	√		

Description of the content (CO wise):

CO.1. To understand knowledge of fundamental principles and basic laws of modelling, MATHEMATICAL MODELING OF CHEMICAL ENGINEERING SYSTEM, Principle of formulations, Mathematical consistency of model, Continuity, equations, Component Y, continuity equations, Energy equations, Equations of motion, Transport equations, Equilibrium, Chemical Kinetics with examples.

CO.2. To understand the approach for mass/heat transfer & CRE, MODELING OF CHEMICAL KINETICS AND REACTOR DESIGN, Modeling for different reaction scheme, Introduction to Reactor Design, Fundamentalsfor Ideal Systems, Introduction A General Approach Ideal , Isothermal Reactors, Numerical Methods for Reactor Systems Design, Reversible Series Reactions, The Semibatch Reactor, Continuous Flow, Stirred Tank Reactor (CFSTR), Multi-Stage Continuous Flow Stirred TankReactor, Equal Size CFSTR in Series

CO.3. To apply the knowledge of differential equations

CO.4. To understand the approach for modelling

CO.5. Formulation of mathematical model for various chemical Engg. system

Course Description:

APPLICATIONS IN CHEMICAL ENGINEERING SYSTEMS

Series of isothermal, constant holdup CSTR'S, CSTR'S with variable Holdups, two heated tanks, Gas-Phase pressurized CSTS, Nonisothermal CSTS, single component vaporizer, Multicomponent flash Drum, Batch Reactor, Reactor with mass Transfer, idial binary distillation column, multicomponent non-ideal distillation column, batch distillation with holdup. TREATMENT OF EXPERIMENTAL RESULTS, Solve above developed modeling equations using polymath/matlab/c++

Text Books/ Reference Books:

1) Mickley H. S., Sherwood T. S., Reed C. E., Application of Mathematical Modeling in Chemical Engineering ,Tata-McGraw-Hill, New Delhi, 2002.
2) Jensen V.G., Jeffrey's G.V., Mathematical Methods in Chemical Engineering", 2nd Ed. Academic Press, London, 1978.
3) Lubyen W. L., Process Modeling, Simulation and Control for Chemical Engineers, McGraw-Hill,New York, 1989.
4) A. Kayode Coker, Modelling of Chemical Kinetics and Reactor Design, Gulf professional publication
5) Incropera F.P. Fundamentals of Heat and Mass Transfer 5th Edition Wiley India Pvt.Ltd Ltd., 2008

Course Code : CML 303
Course Title : Process Control and Instrumentation
Course Credit (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : CML 263
Overlapping/Equivalent courses : None

Course Outcomes/ Objectives:

1) Student shall be able to formulate the dynamic model of different chemical processes.
2) Student shall be able to identify the different measuring instruments and their working principle.
3) Student shall be able to apply analytical tools and design methodologies to tackle process control problems.
4) Student shall be able to apply control schemes along with tuning techniques in chemical processes.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data.
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs → / COs ↓	a	b	c	D	e	F	g	h	i
CO1	**H**	**H**				**M**		**L**	
CO2	**H**	**H**						**L**	
CO3	**H**	**H**				**M**		**L**	
CO4	**H**	**H**				**M**		**L**	

Course Content (CO wise):

Students will be able to analyze the dynamic behavior of processes and will be able to apply control schemes in chemical processes

CO1: Importance, aims and objectives of process control, introduction to system dynamics, concept of dynamic response, first order, second order interacting and non interacting systems, concepts of transfer function, time constant, process gain, overshoot, decay ratio, dead time. Introduction to set point, disturbance, closed loop and open loop control, feedback and feed forward configurations, dynamics of feedback control system.

CO2: Process instruments used for measurement of pressure, temperature, liquid level, flow rate and compositions, pressure gauge, strain gauge, McLeod gauge, vacuum measurement, transducers, transmitters, digital signal processing.

CO3: Types of controllers, P, PI and PID controllers, controller gain, stability analysis, Routh stability criteria. Design of controllers using open loop response, Zigeler – Nichols controller settings, Bode and Nyquist stability criteria.

CO4: Control valve and choice of controller settings. Basic design of pneumatic controllers, electric / electronic controllers, discontinuous control modes – two position, classical and modern control actions. Introduction to set point, error, accuracy, sensitivity, Application of control systems to chemical process equipments such as chemical reactors, heat exchangers, distillation columns, boilers etc.

Textbooks:

1) Donald R. Coughanowr and Kappel, “Process system Analysis & Control”, McGraw Hill Book Company.

2) R.P.Vyas, “An Introduction to Process dynamics & control, Central publication, Nagpur, 2001.

Reference Books:

1) T. Marlin, "Process Control", McGraw Hill, 1995.

2) W.L.Luyben, "Process Modelling Simulation and Control for Chemical Engineers", McGraw Hill, 1990.

3) G.Stephanopoulos, "Chemical Process Control: An Introduction to Theory and Practice", Prentice-Hall, New Jersey, 1984.

Course Code : CML304
Course Title : Chemical Reaction Engineering II
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Outcome/Objective:

Objective of this subject is to expose students to understand the basic reactor analysis catalysis, flow patterns in reactors and its application to chemical engineering

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data.
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Course content (CO wise):

Overview of Chemical Reaction Engineering: Summary of Chemical Reaction Engineering-I, Choosing the right kind of reactor.

Flow Pattern, Contacting, and Non-Ideal Flow: Non ideal flow in reactors, RTD of fluid in reactors, Age distribution, F curve, C curve and E curve, Compartment model, Dispersion model, Tank in Series model.

Introduction to Heterogeneous Reactions: Examples of heterogeneous reactions, contacting pattern and flow modeling.

Solid Catalysed Reactions: Introduction and Spectrum of kinetic regimes, Surface kinetics and rate equation, pore diffusion, porous catalyst, Heat effects, Performance Equation, Experimental methods and rate equation, Controlling Resistance, Product distribution in multiple reactions.

Introduction to Catalyst and Catalytic Reactors: Typical Catalysts, Catalyst Characterizations, Catalyst Deactivation and Regeneration, Packed bed reactor, Fixed Bed, Fluid Bed, Trickle bed, Slurry Reactors etc.

Kinetics and Design of Fluid- Fluid Reactions: The rate equation, Kinetic regimes for mass transfer and reaction, Fast reaction, Intermediate reaction, Slow Reactions, Factors to select the contactor, Straight mass transfer, Various cases of mass transfer with chemical reaction, reaction kinetics.

Kinetics and Design of Fluid- Particle Reactions: Various models for fluid-solid reactions, Shrinking core model, Rate of reaction, Reaction/Mass transfer Control, Rate controlling steps, plug flow and mixed flow of solids.

Textbooks:

1) Octave Levenspiel, “Chemical Reaction Engineering” John Wiley & Sons, Singapore, 1998 3rd Edition.

References:

1) Fogler H.S., “Elements of Chemical Reaction Engineering”, Prentice-Hall, NJ, 2006, 4th Edition.

2) G. F. Froment and K. B. Bischoff, “Chemical Reactor Analysis”, John Wiley & Sons, Singapore, 1990 2nd Edition.

3) Smith J. M., “Chemical Engineering Kinetics”, McGraw Hill, N Y, 1981, 3rd Edition.

Course Code : CML 353
Course Title : Chemical Process Equipment Design
Credits : 3 (3-0-0)
Pre-requisites : General Metallurgy, Mechanical Operations, Strength of materials
Overlaps with : None

Objectives :
After completion of this course the students will be able to do design of industrial pressure vessel and storage vessel.

Course Outcomes:
1) Provide students with basic understanding equipment design
2) To teach students the design of pressure vessel
3) To teach students the design of storage vessel
4) To teach students to apply the design concepts in practical industrial design problem

Program Outcomes for Chemical Engineering at VNIT:
a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs → / COs ↓	a	b	c	d	e	f	g	h	i
CO1	H	H	H						
CO2		H	M						
CO3		H	M						
CO4		H	H			M	M	H	M
Overall	H	H	H			M	M	H	M

Course Content:

Unit-1

Importance of chemical process equipment design, design procedure for pressure vessels subjected to internalpressure, and combined loading, closures for pressure vessels, Code and standards for pressure vessels(IS:2825:1969), materials of construction, selection of corrosion allowance and weld joint efficiency.

Unit-2

Design of pressure vessels subjected to high pressure, monoblock construction, shrink fit construction, externalpressure, optimum proportions of pressure vessels, optimum sizing of vessels.

Unit-3

Design of supports, flanges, nozzles for vessels, Design of jackets (as per IS 2825), coils for pressure vessels.

Unit-4

Mechanical design of storage tanks for volatile and non-volatile liquids, roof and bottom design, optimumproportions of storage tank, storage tanks for solids and its design procedure, Design of cylindrical storagevessel as per IS:803 and rectangular tanks as per IS:804.

Unit-5

Codes and standards for heat exchangers; Baffles; Tie-rods; Tube joining methods; Design of shell and tubeheat exchangers as per IS : 4503 and TEMA standards; design of single effect evaporator

Unit-6

Design of distillation column, absorption column, and reactors

Text Books/ Reference Books:

1) Process equipment dsign-vessel design by Lloyd E. Brownell and Edwin Young, John Wiley, NewYork1963.
2) Chemical Engineering Volume 6 – Design by J.M. Coulson, J.F. Richardson and R. K. Sinnott, Pergamonpress International Edition 1989.
3) Introduction to chemical equipment design – Mechanical Aspects by B.C. Bhattacharyya, CBS Publications.
4) Process Equipment Design by M.V. Joshi and V.V. Mahajani Macmillan India
5) Pressure Vessel Hand book by Eugene F. Megyesy, Pressure vessel company USA.
6) Design of machine elements by V.B. Bhandari, McGraw Hill.
7) Appropriate ISI Specifications and codes for unfired pressure vessels, viz IS: 2825, IS: 803, IS: 804,IS: 1182,IS: 4853, IS: 3658, IS: 3703, IS: 3664, IS: 4260, IS: 4072, IS4503.

Course Code : CMP 304
Course Title : Chemical Reaction Engineering
Course Credits (L-T-P) : 1 credits (0-0-2)
Pre-requisites : CML223- Chemical Reaction Engineering -I
Overlaps with : None

Course Outcomes:

1) Students will understand the basic of chemical engineering and its practical application.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Objectives:

To understand the basic principle of chemical reaction engineering

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
CO1	L	L	H	L	M	L			

Course description:

This course mainly deals with the understanding the basic fundamental principles of chemical reaction engineering by performing different experiments.

Course Content:

CO 1:

1) Three CSTRS Connected in Series

Aim: Study the kinetics of reaction for all the combination for given three CSTR in series

2) PFR & CSTR in Series

Aim: Study the kinetics of reaction for all the combination for given PFR and CSTR in series

3) Isothermal Continuously Stirred Tank Reactor

Aim: To Study the performance of isothermal continuous stirred tank reactor for the reaction ethyl acetate and NaOH

4) Isothermal Plug Flow Reactor

Aim: To Study the performance of isothermal continuous stirred tank reactor for the reaction ethyl acetate and NaOH

5) R.T.D. Studies in Plug Flow Reactor

Aim: To plot the F-Curve and C- Curve for given Plug Flow Reactor

6) Semi Bath Reactor

Aim: To determine overall order of Reactions for bimolecular reactions

7) R.T.D. Studies in Series & Parallel CSTR

Aim: To plot the F-Curve and C- Curve for given Plug Flow Reactor

8) Adiabatic Batch Reactor

Aim: To study the kinetics of reaction adiabatically

9) Isothermal Batch Receiver

Aim: To find the Arrhenius constant

10) R.T.D. Studies in Packed Bed Reactor

Aim: To plot the F-Curve and C- Curve for given packed bed reactor

11) Condensation Polymerization Reactor

Aim: To study the polymerization reaction in given condensation polymerization reactor

12) Fluidized Bed Reactor

Aim: To study the performance of fluidized bed reactor.

Textbooks:

1) Chemical Reaction Engineering, Octave Levenspiel, John Wiley & Sons, Singapore, 1998 3rd Edition

Reference Books:

1) Elements of Chemical Reaction Engineering, Fogler H.S., Prentice-Hall, NJ, 2006, 4th Edition
2) Chemical Engineering Kinetics, Smith J. M., McGraw Hill, N Y, 1981, 3rd Edition.

Course Code : CMP321
Title of the course : Mass Transfer I
Course Credits (L-T-P) : 1 (0-0-2)
Course Type : Core
Pre-requisites : CML221-Mass Transfer I

Course Content/List of Experiments:

1. To determine the Mass Transfer coefficient for Absorption of CO_2 in NaOH solution in packed Column.
2. Study of adsorption of acetic acid on activated charcoal [To verify adsorption isotherms].
3. To determine the number of Heat Transfer Units (HTU) & height equivalent to Theoretical plate (HETP) of Packed distillation column.
4. To study the drying characteristics curve under constant drying condition in rotary vacuum or tray dryer.
5. Diffusion (Liquid – Liquid) –To calculate the diffusion coefficient of vapour in still Air.
6. To study the characteristics of Boiling point diagram.
7. To study the characteristics Cooling Tower experiment.
8. Experiments on Differential Distillation.
9. To determine rate of distillation by Steam Distillation.
10. Performance evaluation of fluid bed dryer.
11. Study of factors affecting rate of Evaporation :-

i) Effect of Surface Area.
ii) Effect of Temperature.

12. Solid liquid extraction
13. Liquid – Liquid Extraction– To determine Overall efficiency for a three stage counter-current and cross current system.
14. Diffusion (Liquid–Air):- To find the diffusion coefficient of vapour instill air.
15. Experiments on Fractional Crystallization.
16. Spray Column Dryer:- To study the Design and operating Principles of Spray Dryer.
17. Plate Column Distillation :- to study the Performance of a rectification column.
18. Determination of Rate of drying, Free moisture content and bound moisture content.

Total Experiments to be conducted / designed: **8-10**

Textbooks:

1. Mass Transfer Operations, Treybal R.E., McGraw Hill Book Co., New York 1980, 3^{rd} Edition
2. Chemical Engineering Vol. I, II & III, Coulson J.M. and Richardson J.F., Pergamon Press, New York 1977

3. Unit Operations of Chemical Engineering, McCabe W.L. and Smith J.C. & Harriot, McGraw Hill Book Co., New York 1980, 5th Edition
4. Principles of Mass Transfer and Separation Process, Binay K. Dutta, PHI Learning Pvt. Ltd., New Delhi, Eastern Economy Edition
5. Introduction to Chemical Engineering, Badger W.L. and Banchero J.T., Tata McGraw Hill Book Co.

Course Code : CMP453
Course Title : Design Lab I
Course Credits (L-T-P) : 1 Credtis (0-0-2)
Pre-requisites **:** None
Overlapping/Equivalent course **:** None

Objective: Objective of this subject is to expose students to understand and drawing the basic symbolic representation and its application to chemical engineering

Syllabus: Symbols used in Drawing, Design and Drawing of various chemical equipments and accessories like storage tank, Jacketed vessel, reaction vessel, hanges coil, gasket, pressure vessel, supports, Agitator, etc.

Minimum 10-12 Imperial size sheets (A-1) covering the above syllabus should be drawn out of which 1/3rd should be drawn using computer software like AutoCAD.

TEXT BOOKS:

1. Joshi M.V., Mahajan V.V., "Process Equipment Design" ,MacMillan India Ltd

REFERENCES

1) Khurmi R.S ,Gupta J.M., "A text book of machine design", S.Chand &Company Ltd, NewDelhi.
2) Dawande S.D., "Process Design of Equipments", Central Techno Publication, Nagpur.

Course Code : CML232
Course Title : Introduction to computing softwares for chemical engineering
Course Credits (L-T-P) : 3 (3-0-0)
Pre-requisites : None
Overlaps with : None
Course Outcomes/ Objectives:

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs → COs ↓	**a**	**B**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**
CO1											
CO2											
CO3											
CO4											
Overall											

Content (CO wise):

CO1: Introduction to Softwares- Documentation; Development Environment; Desktop Tools, Other development environment tools.

CO2: Manipulating Matrices-Matrices and Magic squares, Expressions, Working with Matrices, More about Matrices and Arrays, Controlling commnad window input and output; Graphics-Basic plotting, Editing plots, Mesh and surface plots, Printing and Handling the Graphics.Programming with MATLAB-Flow control, Other data structures, Scripts and Functions.

CO3: Examples of Chemical engineering solved problems using MATLAB-Equations of state, Vapour liquid equilibrium,

CO4: Chemical reaction equilibria, Reaction-kinetic system, Transport processes, etc

Textbooks:

1) W. Fred Ramirez, Computational Methods in Process Simulation, Elsevier Science & Technology Books

Reference Books:

1) Andrew Knight, Basics of MATLAB and Beyond, Chapman & Hall/CRC press LLC., 2000
RudraPratap, Getting Started with MATLAB7: A Quick Introduction for Scientists and Engineers, Oxford University Press, Newdelhi, India.

Course Code : CML 383
Title of the Course : Advance Heat Transfer
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : Heat Transfer
Overlapping/Equivalent courses : None

Course Objectives/Outcomes:

1) Design a given heat transfer equipment including pressure drop calculations for a given fluid, flow rate and heat duty.
2) To appreciate the construction features of various heat transfer equipment and choose among them, the best suited for a specific job
3) Construct a heat exchanger network for a given set of hot and cold streams for optimal energy usage
4) Determine whether a given heat exchanger can perform the desired duty using NTU method

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

POs → / COs ↓	a	b	c	d	e	f	g	h	i
CO1	H	H	H						
CO2		M							
CO3	H	H	H						
CO4	H	H	H						

Course content (CO wise):

CO.1. Classification of heat exchangers, recuperative, regenerative and direct contact type, double pipe heat exchangers, co-current counter, current flow arrangement, overall heat transfer coefficient.

CO.2. Fixed tube sheet, floated head and U–tube shell and tube heat exchangers, their design procedures, number of passes in heat exchangers, fouling of heat exchangers, baffles in heat exchangers, selection of heating and cooling media for heat exchangers, Troubleshooting of shell and tube heat exchangers, thermal stresses and vibrations in shell and tube heat exchangers.
CO.3. Plate heat exchangers, design procedure, advantages over shell and tube heat exchangers, spiral plate heat exchangers, helical coil heat exchangers.
CO.4. Heat Regenerators, fixed and fluidized bed, Evaporators types and their operational characteristics. Single stage and multistage evaporation system, Steam economy, boiling point rise of solution and its effect on evaporation system, rising film and falling film evaporators.
CO.5. Effectiveness of heat exchanges, NTU method. Heat Transfer in jacketed vessels, boilers, furnaces and reactors, reboilers, heat transfer in agitated vessels with and without coils
CO.6. Pinch Technology, Composite enthalpy curves, Heat Integration and Network Design, Pinch Temperature Determination, Design above and below the pinch

Text Books/ Reference Books:

1) Sinnot, R., Towler ,G., Chemical Engineering Design, Volume 6, Fifth Edition, Elsevier, New Delhi.
2) Dutta, Binay K, Heat Transfer: Principles and Applications, PHI Learning Private Limited, 2001.

Course No. : CML524
Course Title : Polymer Science and Engineering
Course Credits (L-T-P) **: 3 (**3-0-0)
Pre-requisites : None
Overlaps with : None

Objectives: To understand the different types of plastics, To understand the processing of plastics.

Course Outcomes

1) To understand the preparation and testing of plastics.
2) To understand the factors affecting properties of polymers.
3) To understand the fundamentals of waste management.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a
h. global and societal context
i. A recognition of the need for, and an ability to engage in lifelong learning
j. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	h	I
CO1	H	H	H				M	M	M
CO2	H	H	H				M	M	M
CO3	H	H	H				M	M	M

Course Content:

CO -1: To understand the different types of plastics.

Comparison of thermoplastics and thermoset plastics; Thermoset plastics - Types of resins, Interpenetrating Polymer Networks (IPN); Thermoplastics - Types of aliphatic and aromatic thermo plastics, copolymers, Blends and alloys; Liquid crystal plastics; cellular plastics; oriented plastic materials.

To understand the processing of plastics.

Processing: Basics of process design, Classification & general aspects of processes - molding & forming operations, Post die processing; Decoration of plastics - Printing, Vacuum Metalizing, In-mold decoration. Additives & Compounding - Different types of additives, Batch mixers, continuous mixers, Dispersive and distributive mixing, Characterization of mixed state.

Fundamentals on Viscous & Viscoelastic behavior of polymer melt, Rheological measurements and Polymer processability. Non isothermal aspects - Temperature effect on rheological properties, Crystallization, Morphology & Orientation, plastic memory, Molecular weight effects on processing and properties.

CO-2: To understand the preparation and testing of plastics.

Properties & Testing of plastics: Basic concepts of testing, National & International standards, Test specimen preparation, Pre conditioning & Test atmosphere.

Identification of plastics by simple test - Visual examination, Density, Melting point, Solubility test, Flame test, Chemical tests.

To understand the factors affecting properties of polymers.

Effect of shape & structure on material properties, Long - term & short - term mechanical properties, crazing, Permeability & barrier properties, Environmental-stress cracking, Melt flow index, Heat deflection temperature, Vicat softening temperature, Glass transition temperature, thermal conductivity, Co-efficient of thermal expansion, Shrinkage, Thermal stability, Flammability.

CO-3: To understand the fundamentals of waste management.

Waste management & Recycling: Plastics waste and the associated problems, Integrated waste management - source reduction, recycling & sustainability correlation, energy recovering process. Environmental issues, policies and legislation in India.

Kinetics of polymerization (addition and condensation)

Text Books:

1) Plastics - How Structure Determines Properties, Gruenwald G, Hanser Publishers, 1993
2) Polymer Processing Principles and Design, Baird D. G. and Collias D. I., Butterworth-Heinemann, 1995
3) Hand Book of Plastics Testing Technology, Vishu Shah, John Wiley & Sons Inc. New York
4) How to identify Plastics by Simple Methods.J.S.Anand, K.Ramamurthy, K.Palanivelu, CIPET, Chennai, 2nd edition
5) Plastics and the Environment, Anthony L. Andrady (Ed.), Wiley Interscience, New York.

- **Chemical Informatics**
- **Course Code:** CML 389
- **Course Type:**Elective
- **Credits:** 03, Class Schedule: 3 Classes per week (55 minutes each)

I. Course Description

This course mainly focuses on providing the knowledge of various cheminformatics tools available that can be utilized for the applications such as pharmaceutical drug design and discovery as well as material design and development.

II. Pre-requisites

- None

III. Course Objectives

The course aims:

1. To introduce cheminformatics concepts for handling the molecular data and prediction of properties.
2. To make use of cheminformatics tools for the discovery, design and development of novel chemicals, drugs and materials.

IV. Course Outcomes (COs)

Upon completing this course, students will be able to:

1. **CO1:**To learn and utilize cheminformatics tools for pharmaceutical drug design and development.
2. **CO2:**To learn and utilize cheminformatics tools for the design and development of novel materials and chemicals
3. **CO3:**To learn chemical structure representation and ways to handle it on computers

V. Expanded Course Description

Unit I: Cheminformatics Introduction (6 Hours)

- Introduction to cheminformatics, History and evolution of cheminformatics
- Use of cheminformatics, Prospects of cheminformatics, Molecular modeling and structure elucidation

Unit II: Representation of Molecules and Chemical Reactions (6 Hours)

- Nomenclature; Different types of notations; SMILES coding; Matrix representations; Structure of Molfiles and Sdfiles; Libraries and tookits; Different electronics effects; reaction classification
- Representing 2D and 3D structures, Kinds of 2D structures representation; atom lookup and connection tables; graph theory; SMILES; SD files; representation nuances; descriptors

Unit III: Database screening bySimilarity Search (4 Hours)

- Database screening by similarity search: Database concepts, Design of chemical databases
- Structure search, Pharmacophore search, Fingerprints.

Unit IV: Molecular Mechanics (6 Hours)

- Calculation of physical and chemical data, molecular mechanics
- Descriptors for chemical compounds, Artificial intelligence systems in chemical engineering
- Methods for data analysis, Chemical markup language (CML)

Unit V: Prediction of Physical and Chemical Properties(8 Hours)

- Prediction of physical and chemical properties, Structure–Activity correlations, Regression methods
- Chemical reactions and synthesis design, Elements of bioinformatics and genomics, Docking

Unit VI: Drug Design (6 Hours)

- Drug design
- Absorption-Distribution-Metabolism-Excretion (ADME) Pharmacokinetics
- Drug delivery

VI. Textbooks & References

1. Andrew Bender, Jonathan M Goodman, Cheminformatics, Oxford University Press-2007
2. Gasteiger J. and Engel T., Chemoinformatics, A Text Book, Wiley VCH.
3. Stuart Schreiber, Tarun M. Kapoor, Chemical Biology: From Small Molecules to Systems Biology and Drug Design Chemical Release 2001.
4. Leach A. R. and Gillet V. J., An Introduction to Cheminformatics, Revised edition, Springer, 2007.
5. Karthikeyan M., Vyas R., Practical Cheminformatics, Springer, 2014.

VII. Evaluation Scheme

- **Class Tests and Assignments:** 20%
- **Mid-Semester Exam (1.5 Hours):**25%
- **End-Semester Exam (3 Hours):** 55%

VIII. Relationship Between Course Outcomes (COs) and Program Outcomes (POs)

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	1	2	-	-	-	2	2	-	1
CO2	2	2	3	1	2	-	1	-	2	2	-	1
CO3	1	1	1	-	2	-	-	-	-	-	-	

Key:

3 = Strong correlation, 2 = Moderate correlation, 1 = Weak correlation, - = No correlation

IX. Consultation Hours

Chamber consultation hours will be announced in class.

- **Space Technology for Chemical Engineers**
- **Course Code:** CML 4XX
- **Course Type:** Elective
- **Credits:** 03, Class Schedule: 3 Classes per week (55 minutes each)

I. Course Description

This course covers the application of chemical engineering principles to space exploration, propulsion systems and materials science for space applications.

II. Pre-requisites

- None

III. Course Objectives

The course aims:

3. To introduce chemical engineering principles to space technology
4. Toidentify innovative materials and solutions for space exploration challenges.

IV. Course Outcomes (COs)

Upon completing this course, students will be able to:

4. **CO1:**Tounderstand chemical engineering principles related to space technology
5. **CO2:**To develop innovative solutions for space exploration challenges
6. **CO3:**To identify and design newer chemicals and materials for space applications

V. Expanded Course Description

Unit I: Introduction to Space Technology (6 Hours)

- History of space exploration, Space organizations, Space missions and applications,
- Relevance to chemical engineering

Unit II: Propulsion systems (6 Hours)

- Propellants (fuel and oxidizer)
- Chemical propulsion, Electric propulsion, Nuclear propulsion
- Solar sails

Unit III: Material Science (6 Hours)

- Nanomaterials for space applications, Materials selection for spacecraft design
- Thermal protection systems, Smart materials

Unit IV: Energy generation and storageConcepts (6 Hours)

- Spacecraft power systems

- Solar power generation, Nuclear power generation
- Fuel cells and battery systems

Unit V: Life support systems(6 Hours)

- Oxygen generation, Carbon dioxide removal
- Water recycling and purification, Food production and processing,
- Waste management

Unit VI: Space exploration(6 Hours)

- Space exploration
- Planetary resource extraction and utilization
- Use of artificial intelligence and robotics
- Space policies and regulation

VI. Textbooks & References

1. "Space Technology: A short introduction" Springer 2023.
2. "Future Spacecraft Propulsion Systems: Enabling Technologies for Space Exploration" Paul A. Czysz and Claudio Bruno, Springer 2006.
3. "Solid Rocket Propulsion Technology" by A. Davenas, Pergamon Press, 1993.
4. "Space Material Sciences" by A I Feonychev 2011.

VII. Evaluation Scheme

- **Class Tests and Assignments:** 20%
- **Mid-Semester Exam (1.5 Hours):**25%
- **End-Semester Exam (3 Hours):** 55%

VIII. Relationship Between Course Outcomes (COs) and Program Outcomes (POs)

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	-	2	2	2	1	2	-	1
CO3	3	1	3	3	1	-	2	2	1	2	-	1

Key:

3 = Strong correlation, 2 = Moderate correlation, 1 = Weak correlation, - = No correlation

IX. Consultation Hours

Chamber consultation hours will be announced in class.

Course Code : CML 351
Course Title : Chemical Technology
Course Credits (L-T-P) : 03 (3-0-0)
Pre-requisites : None
Overlaps with : None

Course description: The course deals with the preparation of chemical process flow diagrams involving various unit operations and unit processes in synthesis of organic and inorganic products such as petroleum products, coal chemicals, fertilizers, acids, cement, sugar, alcohol, glass, paper and pulp, industrial gases, chloro-alkali chemicals etc.

Course Objectives:

1) To study the overall process flow in production of chemical products of importance
2) To gain knowledge of various unit operation and processes involved in the production process
3) To gain knowledge in performing overall mass and energy balance of a process
4) To attain the knowledge of flow/ separation pattern of material
5) To understand the approach for modeling of a process plant

Course Outcomes (CO):

At the end of the course, the student Students will gain knowledge on

1) Basic unit process and unit operations involved in production of various chemicals.
2) Raw materials, reactions, catalysts, temperature, pressure and time of operation of various unit operations and process involved in production of various organic and inorganic chemicals.
3) Production of various intermediate products which serves as main raw materials in manufacture ofa specific product
4) Preparation of process flow diagrams involved in production of various chemicals

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO-1	?	?				?						
CO-2	?		?									
CO-3	?	?	?			?			?			
CO-4	?	?	?				?		?	?		?

Course Content (CO wise):

Introduction and overview of Chemical Process Technology. Preparation of process flow diagrams, Instrumentation diagrams and Process symbols.

Petroleum refinery processes: Introduction to crude oil, Crude refining processes (i) physical processes (Desalting/dehydration, Crude distillation, Propane deasphalting Solvent extraction and dewaxing, Blending, (ii) Chemical process (thermal process –Visbreaking, Delayed coking, Flexicoking), Catalytic Processes –Hydrotreating, Catalytic reforming, Catalytic cracking, Hydrocracking, Catalytic dewaxing, Alkylation, Polymerization, Isomerization

Petrochemical Industries: production of petrochemical feedstocks, olefins and aromatics, intermediates from olefins and aromatics. Manufacture of ethylene, propylene, butylenes, benzene, toluene etc.

Inorganic Chemical Industries: chloro-alkali industries, manufacture of acids-sulfuric, nitric, phosphoric acids, Fertilizers- ammonia, urea, Ammonium sulfate, ammonium nitrate, Urea, SSP and TSP and miscellaneous fertilizers.

Natural products -manufacture of sugar, starch, and its derivatives, Pulp, Paper, oil and fats, Rayon industries. Edible oils: extraction and refining, fat splitting, soaps and detergents.

Polymerization industries (ethylene, polyethylene, propylene,polypropylene, butylenes, benzene, toluene, PVC and polyester synthetic fibers etc.

Textbooks:

1) Dryden C.E.; Outlines of Chemical Technology: East West Press, 1973.
2) Shreve R.N and Brink J.M; Chemical Process Industries: McGraw Hill Co.,New York, 1977.
3) P. H. Groggins, Unit Processes in Organic Synthesis, 5th Ed., McGraw Hill, 1984.
4) Soni P.L. and Kalyal; Textbook of Inorganic Chemistry, 20th Edition, S. Chand &Co., New Delhi.
5) Venkateshwarulu D.; Manual of Chemical Technology: Vols I and II, IIT, Madras,8. 1977.

Course Code : CML 352
Course Title : Transport Phenomena
Course Credits (L-T-P) : 4 Credits (3-1-0)
Pre-requisites : CML202, CML221, CML222.
Overlaps with : None

Course Objectives:

Students will be able to develop and use velocity profile, flux profile model equations for different systems in heat, mass and momentum transfer using shell balance approach

Course Outcomes:

1) Correlate the heat mass and momentum transfer
2) Explain the shell balance approach and boundary conditions
3) Obtain velocity, concentration and temperature profile, heat mass and momentum flux profile and other important equation.
4) Solve design problem using profile equations

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	H					H	M	L	L
2	H	H				M			
3	H	H	M			M			L
4	H	H	M					L	

Course description:

This course mainly deals with application of knowledge gain by students in fluid mechanics, heat transfer and mass transfer. This subject also covers the topic which shows the similarity between fluid mechanics, heat transfer and mass transfer

Course Content (CO wise):

CO 1: Similarity between heat, momentum and mass transport and mass transport Molecular and Convective Transport, Definition of Transport Properties, Viscosity, Thermal Conductivity, Diffusivity, Newton's Law of Viscosity, Fourier's Law of Heat Conduction, Fick's First Law of Diffusion, Dimensionless Numbers in Molecular Transport, Convective transport and total flux.
Inter-phase Transport and Transport Coefficients, Physical Interpretation of Friction Factor, Heat Transfer Coefficient, Mass Transfer Coefficient, Dimensionless Numbers in inter-phase Transport. Various Transport Analogies, The Reynolds Analogy, The Chilton-Colburn Analogy Evaluation of transfer coefficient: Engineering Correlations

CO 2: Shell Momentum Balances and Velocity Distributions in Laminar Flow. Shell Momentum Balances and Boundary condition. Different cases for discussing shell moment balance approach like, Flow of a Falling Film, Flow Through a Circular Tube, Flow through an Annulus, Flow of Two Adjacent Immiscible Fluids.
Shell Energy Balances and Temperature Distributions in Solids and Laminar Flow. Shell Energy Balances; Boundary Conditions. Different cases for discussing shell energy balance approach like, Heat Conduction with an Electrical Heat Source, Heat Conduction with a Nuclear Heat Source, Heat Conduction with a Viscous Heat Source, Heat Conduction with a Chemical Heat Source, Heat Conduction through Composite Walls, Heat Conduction in a Cooling Fin

CO 3: Shell mass balance for Concentration Distributions in Solids and Laminar Flow. Shell Mass Balances; Boundary Conditions, Diffusion through a Stagnant Gas Film, Diffusion with a Heterogeneous Chemical
Reaction, Diffusion with a Homogeneous Chemical Reaction, Diffusion into a Falling Liquid Film (Gas Absorption), Diffusion into a Falling Liquid Film (Solid Dissolution), Diffusion and Chemical Reaction inside a Porous Catalyst, Diffusion in a Three-Component Gas System

CO 4: The Equations of Change for Isothermal Systems
The Equation of Continuity, The Equation of Motion, The Equation of Mechanical Energy, The Equation of Angular Momentum, The Equations of Change in Terms of the Substantial Derivative, Use of the Equations of Change to Solve Flow, Steady Flow in a Long Circular pipe, Falling Film with Variable Viscosity, Operation of a Couette Viscometer, Shape of the Surface of a Rotating Liquid, Flow near a Slowly Rotating Sphere, Dimensional Analysis of the Equations of Change

Textbooks:

1) R.B. Bird, W. Stewart and E.N Lightfoot, Transport Phenomena, John Wiley & Sons, 2nd Edition, 2006

2) Ismail Tosun, Modeling in Transport Phenamena, Elsevier Science & Technology Books, 2^{nd} Edition, 2007

Reference Books:

1) C.O. Bennet and J.E. Myres, Momentum, Heat & Mass Transfer, McGraw Hills, 3^{rd} Editon, 1994

2) R. Brodkey and H. C. Hershey, Transport Phenomena – A Unified Approach, volume 1, McGraw Hill Book Co., 2^{nd} Edition, 1988

3) C.J. Geankoplis, Transport Processes and Separation Process Principles, Prentics Hall India Ltd., 4^{th} edition, 2003

4) G.S. Laddha and T.E. Degaleesan, Transport Phenomena in Liquid Extraction, Tata McGraw Hill Book Co., 1^{st} edition, 1978

- **Energy and Environment**
- **Course Code:** CML 354
- **Course Type:** Core
- **Credits:** 03, Class Schedule: 3 Classes per week (55 minutes each)

I. Course Description

This course mainly focuses on conventional and non-conventional energy sources such as coal, petroleum, natural gas, nuclear energy, bioenergy, hydropower, geothermal energy, solar energy and wind energy. It is designed to establish a connection between the various energy sources and their environmental influences. Several novel concepts for enhancing the energy production and treating the environmental problems are also addressed in this course.

II. Pre-requisites

- None

III. Course Objectives

The course aims:

5. To investigate various conventional and non-conventional energy sources and their environmental issues.
6. To understand the alteration in the existing technologyfor solving environmental concerns and to optimize the energy production.

IV. Course Outcomes (COs)

Upon completing this course, students will be able to:

7. **CO1:**To analyze and interpret energy production from both conventional and non-conventional resources
8. **CO2:**To optimize the energy production by modification in the existing technology
9. **CO3:**To identify the environmental concerns related to each and every energy sources
10. **CO4:**To resolve the environmental issues by modification in the existing technology

V. Expanded Course Description

Unit I: Introductory Concepts (4 Hours)

- Conventional and non-conventional energy sources; coal, oil and natural gas, solar energy; wind energy; geothermal energy; hydropower; bioenergy; nuclear energy.

- Current and future energy requirements in India and across the world and associated environmental problems.

Unit II: Coal (7 Hours)

- Solid fuels, Coal and coal derived fuels; Characteristics, production methods and uses
- Coal combustion technology, Carbon capture and sequestration

Unit III: Oil and Natural Gas (7 Hours)

- Oil and Natural Gas: Fuels from oil and gases: Characteristics, production methods and uses.
- Oil spills and treatment
- Natural gas hydrates

Unit IV: Solar, Geothermal and Hydro Energy (8 Hours)

- Solar Energy: Solar energy utilization, Thermal application and photovoltaic applications
- Geothermal energy
- Hydro energy utilization, Environmental aspects

Unit V: Wind and Bio Energy (6 Hours)

- Wind Energy: Power extraction by wind turbine, Betz criterion;
- Bio Energy: Biomass conversion for fuels; production methods based on thermochemical and bioconversion. Characteristics and uses, Carbon neutrality

Unit VI: Nuclear Energy (4 Hours)

- Nuclear Energy: Nuclear fission fuels processing, nuclear reactions and nuclear reactors, nuclear engineering, Environmental aspects.

VI. Textbooks & References

1. G.N. Rai, "Non conventional energy sources," Khanna Publishers, New Delhi.
2. Samir Sarkar, "Fuels and Combustion", 2nd Edn, Orient Longman Publication, 1988.
3. J. G. Speight, Fuel Science & Technology Handbook, Dekker, 1990.
4. V.Balzani and N. Armaroli, Energy sustainable world, Wiley-VCH, 2011
5. G. Boyle, Renewable Energy, Oxford University Press, 2004.

VII. Evaluation Scheme

- **Class Tests and Assignments:** 20%
- **Mid-Semester Exam (1.5 Hours):**25%

- **End-Semester Exam (3 Hours):** 55%

VIII. Relationship Between Course Outcomes (COs) and Program Outcomes (POs)

COs/POs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	-	2	-	-	-	-	-	-	-	1
CO2	3	1	3	2	-	-	-	-	1	2	-	1
CO3	3	2	-	2	-	3	3	2	-	-	-	2
CO4	3	1	3	2	1	2	3	-	1	2	-	1

Key:

3 = Strong correlation, 2 = Moderate correlation, 1 = Weak correlation, - = No correlation

IX. Consultation Hours

Chamber consultation hours will be announced in class.

Course Code : CMP 302
Course Title : Chemical Process Modelling and Simulation
Credits : 1 (0-0-2)
Pre-requisites : CML 302
Overlap with : None

Course Outcomes:

1) To understand knowledge of fundamental principles and basic laws of modeling
2) To understand the approach for mass/heat transfer & CRE
3) To apply the knowledge of differential equations
4) To understand the approach for modeling

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	h	i
CO1	√		√			√			
CO2	√		√			√			
CO3	√		√			√			
CO4	√		√			√			

Course Content (CO Wise):

The following experiments have to be conducted using any one software Polymath/C / C++/ Fortran Depending on availability on machine (Any six out of 10 listed below).

1. Gravity Flow tank.
2. Three CSTR's in series – open loop.
3. Three CSTR's in series – closed loop.
4. Non-isothermal CSTR.
5. Complex reaction scheme (Batch Reactor)

6. Second order complex batch reactor
7. Series parallel reaction scheme
8. Semi-batch reactor model
9. Complex reaction model
10. Parallel second order reaction scheme
11. Reversible and irreversible 1st order reactions
12. 2nd order series reactions
13. Complex set of series parallel reactions
14. 10. Equal Size CFSTR In Series

Text Books/ Reference Books

1) Mickley H. S., Sherwood T. S., Reed C. E., Application of Mathematical Modeling in Chemical Engineering ,Tata-McGraw-Hill, New Delhi, 2002.
2) Jensen V.G., Jeffrey’s G.V., Mathematical Methods in Chemical Engineering”, 2nd Ed. Academic Press, London, 1978.
3) Lubyen W. L., Process Modeling, Simulation and Control for Chemical Engineers, McGraw-Hill,New York, 1989.
4) A. Kayode Coker, Modelling of Chemical Kinetics and Reactor Design, Gulf professional publication
5) Incropera F.P. Fundamentals of Heat and Mass Transfer 5th Edition Wiley India Pvt.Ltd Ltd.,2008

Course Code : CMP303
Title of the course : Process Control and Instrumentation
Course Credits (L-T-P) : 1 (0-0-2)
Course Type : Core
Pre-requisites : CML303

Objectives:

1. To make students aware of working of Different process control instruments through hands on training.
2. To make students to correlate theory and practical process control through principles, fundamental concepts and by experimentation.

Course Outcomes:

1) Students will be able to apply the knowledge of control theory for understanding the various processes, carried out in the Chemical process industry.
2) Students demonstrate their ability of understanding the process control and its application by virtue of experimentation.
3) Students will be able to learn due care and precautions in handling measuring instruments.

Expanded Course description:

Sr. No	**List Of Experiments**
1.	**Interacting System**
2.	**Non-Interacting System**
3.	**Flapper Nozzle Trainer**
4.	**Response Of U-Tube Manometer**
5.	**Control Valve Characteristics**
6.	**Water Temperature Control System Trainer**
7.	**Level Measurement By Air Purge Method**
8.	**Determination of Time Constant of Thermometer And Thermocouple**
9.	**Study of Temperature Measurement Using Sensor RTD**
10.	**Study of Multi Process (Cascade) Control System**

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams

g. The broad education necessary to understand the impact of engineering solutions in a global and societal context

h. Recognition of the need for, and an ability to engage in lifelong learning.

i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

Course Objectives	**Program Outcomes**								
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
CO1	**H**				**H**	**H**		**L**	
CO2	**H**							**L**	
CO3					**L**	**L**			
Overall	**H**				**M**	**M**		**L**	

Textbooks/Reference books:

1) T. Marlin, "Process Control", McGraw Hill, 1995.
2) W.L.Luyben, "Process Modelling Simulation and Control for Chemical Engineers", McGraw Hill, 1990.

G.Stephanopoulos, "Chemical Process Control: An Introduction to Theory and Practice", Prentice-Hall, New Jersey, 1984.

Course Code : CML 403
Course Title : Plant Design and Economics
Course Credit (L-T-P) : 3 credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Outcomes/ Objectives:

After the course students will be able:

1) To read complex P&ID diagrams.
2) To perform an economic and profitability analysis of a proposed chemical plant.
3) To appreciate the importance of safety and incorporate safety features in design.
4) To appreciate that the chemical plants are not always run at design conditions and will experience faults.
5) To write and present a design dissertation.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. Recognition of the need for, and an ability to engage in lifelong learning.
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference)

POs / COs	a	b	c	d	e	F	G	h	i	j	k	L
CO1	M											
CO2	H		M								M	
CO3	H	H	H					H				
CO4	H	M	M									
CO5									H	H		

Course Contents (CO wise):

CO1: Block flow diagram, process flow diagram, piping and instrumentation diagram (P&ID).

CO2: Conceptual design and synthesis of a process flow diagram, development of PFD from generic BFD.

CO3: Plant layout, location and site selection.

CO4: Estimation of capital costs, purchased equipment costs, total capital cost of a plant, bare module cost- base and non-base conditions, estimation of manufacturing costs, cost of labor, utility cost, raw material costs.

CO5: Investment and the time value of money, different types of interest, cash flow diagrams, inflation, Annuities, depreciation, taxation, profitability analysis, criteria for profitability for large projects, net present value, rate of return, evaluation of equipment alternatives, evaluation of risks, sensitivity and uncertainty analysis.

CO6: Safety design: Major chemical industry disasters, safety hierarchy, basic process control safety, Alarms, SIS (safety interlock system), HAZOP, safety valves.

CO7: Operability issues and troubleshooting in a chemical plant.

CO8: Introduction to computer aided flow sheeting.

Textbooks/ References:

1) R Turton, R Balie, W B Whiting, J Shaeiwitz, D Bhattacharya Prentice Hall (4th Edition) Analysis, Synthesis, and Design of Chemical Processes 2013

2) Douglas J McGraw-Hill Sciences (1st Edition) Conceptual Design of Chemical Processes

3) Study material developed by Kevin Dunn and Dr. Thomas Marlin.

Course Code : CMP 401
Course Title : Mass Transfer II
Course Credits (L-T-P) : 1 Credits (0-0-2)
Pre-requisites : Mass Transfer II
Overlapping/Equivalent courses : None

Course Objectives:

The objective of this course is to introduce the mass transfer operations and how to quantify, formulate, and solve engineering problems involving different mass transfer operations like distillation, leaching, liquid-liquid extraction, absorption. To demonstrate that how to apply mass balances and its transfer and analyze systems.

Course Outcomes:

1) To understand the basic concepts and principles of mass transfer operation like distillation, absorption, liquid-liquid extraction, solid-liquid extraction (leaching).
2) How to identify, quantify or formulate, and solve the engineering problems involving mass transfer.
3) Design and analysis of mass transfer processes and equipments.
4) Applications of mass transfer phenomena and its optimum use.

Program Outcomes for Chemical Engineering at VNIT:

- **a.** An ability to identify, formulates, and solves engineering problems
- **b.** An ability to design a system, component, or process to meet desired needs
- **c.** An ability to design and conduct experiments, as well as to analyze and interpret data
- **d.** An ability to communicate effectively
- **e.** An understanding of professional and ethical responsibility
- **f.** An ability to function on multidisciplinary teams
- **g.** The broad education necessary to understand the impact of engineering solutions in a global and societal context
- **h.** A recognition of the need for, and an ability to engage in lifelong learning
- **i.** A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	H	M	L	H		M			
2	H	M	H			L	M		L
3	H	M		L	M	L			
4	H	M	L	M	L	L	M	L	L

Course description:

The course presents the mass transfer operations from a chemical engineering viewpoint. The basic concepts, design and its application regarding the mass transfer operations like Distillation, Absorption, Liquid-liquid extraction and Solid-liquid extraction are stressed in this course.

Course Content (CO wise)

CO1: *Unit I (6 hrs):* Distillation: Vapour – liquid equilibria, Raoult's law, X-Y and H-X-Y diagrams, differential distillation and equilibrium distillation, steam distillation, azeotropic distillation, extractive distillation.

CO2: *Unit II (8 hrs)*

Fractionation, binary distillation, plate and packed columns for distillation, analytical and graphical methods for estimation of number of stages required in distillation column, minimum reflux ratio, optimum reflux ratio, number of stages at optimum reflux, murphree plate efficiency and overall plate efficiency, effect of feed conditions on number of plates for separation.

CO3: *Unit III (4 hrs)*

Concept of HETP, HTU, NTU in distillation, plate and packed columns, packings for packed columns, pressure drop in plate and packed columns, bubble cap, sieve tray, valve tray plate columns.

Unit IV (6 hrs)

Absorption Equilibrium relationships, two film theory, penetration theory, surface renewal rate theory, concept of driving force and mass transfer coefficient, plate column and packed columns for absorption, selection of solvent for absorption and absorbers design procedures.

CO4: *Unit V (6 hrs)*

Liquid – Liquid Extraction fundamentals, selection of solvent for extraction, estimation of mass transfer coefficients, triangular diagram representation, equipments for liquid – liquid extraction, plate and packed columns, spray columns, rotary disc contactors, design procedures and equipment selection criteria. Single stage, multistage operations etc.

Unit VI (6 hrs)

Solid – Liquid Extraction fundamentals, Solvent selection, equilibrium relationship, triangular diagram representation, single stage, multistage concurrent and counter current operation, equipments for solid – liquid extraction, their design procedure and selection criteria.

Textbooks:

1) Treybal R.E., Mass Transfer Operations, McGraw Hill Book Co., New York 1980.
2) Coulson J.M. and Richardson J.F., Chemical Engineering Vol. I, II & III, Pergamon Press, New York 1977.

Reference Books:

1) Badger W.L. and Banchero J.T., Introduction to Chemical Engineering, 4th edition, Tata McGraw Hill Book Co..

2) McCabe W.L. and Smith J.C. & Harriot, Unit Operations of Chemical Engineering, 5th edition, McGraw Hill Book Co., New York 1980.

3) Binay K. Dutta, Principles of Mass Transfer and Separation Process, PHI Learning Pvt. Ltd., New Delhi

Course Code : CMP403
Course Title : Design Lab-II
Course Credits (L-T-P) : 1 Credits (0-0-2)
Pre-requisites : Design lab-I, Chemical Process Equipment Desin
Overlapping/Equivalent courses : None

Course Outcomes/ Objectives:

1) Design and drawing of chemical equipments and flow sheets such as heat exchanger, distillation column, jacketed vessel, agitated vertical column, evaporator, fermentor, reactor cyclone separator, autoclave, dryer, pump, flow sheets etc.
2) To introduce application of design and drawing of chemical equipments

Mapping with Pos:

POs→ COs↓	a	b	c	d	e	f	g	h	i	j	k	l
CO1	H	L	H	L	L	L	M	L	L	M	L	L
CO2	M	H	H	L	L	M	L	L	L	M	L	L
Overall	H	H	H	L	L	M	M	L	L	M	L	L

Course content (CO wise):

- Design and drawing of chemical equipments and flow sheets such as heat exchanger, distillation column, jacketed vessel, agitated vertical column, evaporator, fermentor, reactor cyclone separator, autoclave, dryer, pump, flow sheets etc.
- Preparation of working drawing part list & assembly drawings of plant layouts and piping drawing, device drawing. Minimum 10-12 Imperial size sheets (A-1) covering the above syllabus should be drawn out of which 1/3rd should be drawn using computer software like Auto CAD and any others

Textbooks/ Reference Books:

1) Joshi M.V., Mahajan V.V, Process Equipment Design, Macmillan India Ltd
2) Khurmi R.S ,Gupta J.M. A Text Book Of Machine Design, S.Chand &Company Ltd, New Delhi
3) Dawande S.D. Process Design of Equipments, Central Tecno Publication, Nagpur.

Course Code : CML 393
Course Title : Innovative Design
Course Credits (L-T-P) : 3 (2-0-2)
Pre-requisites : None
Overlaps with : None

Course Objectives:
1. To contribute to the understanding of human creativity
2. To orient students outward, toward the rest of society and to create a broader outlook to face the challenges of the real world effectively
3. To augment engineering education by adding the design component in thinking and doing
4. To impart product and system building knowledge and skills
5. To tackle real life problem by cross-functional and multi-disciplinary team approach

Course Outcomes:
1. will be able to identify and define a problem
2. Learn to frame the design challenge properly
3. Ideate and iterate solutions
4. Develop skills and attitudes such as experimentation, design thinking, teamwork, communication, societal context and business context
5. Learn to participate more fully in society

Program Outcomes for Chemical Engineering at VNIT:
- **a.** An ability to identify, formulates, and solves engineering problems
- **b.** An ability to design a system, component, or process to meet desired needs
- **c.** An ability to design and conduct experiments, as well as to analyze and interpret data
- **d.** An ability to communicate effectively
- **e.** An understanding of professional and ethical responsibility
- **f.** An ability to function on multidisciplinary teams
- **g.** The broad education necessary to understand the impact of engineering solutions in a global and societal context
- **h.** Recognition of the need for, and an ability to engage in lifelong learning.
- **i.** A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	**Program Outcomes (PO)**								
	a.	**b.**	**c.**	**d.**	**e.**	**f.**	**g.**	**h.**	**i.**
CO1.	H	H		H		H	M		H
CO2.		M		H		H	M		H
CO3.		M	H	H		H			
CO4.				H	M	H	H	H	H

CO5.				H		H	H		

Course description:

The course focuses on innovative engineering design in a team-based, cross-disciplinary setting. “Innovative Design” implies both identifying and solving real – world problems for real people. The course engages students with a real problem, which has no given solution in an industrial / social context, develop social and collaborative skills, introduce new product development methods in a project environment.

The course emphasizes the use of Project Based Learning where students participate in active and experiential learning through real product development situations. It will start with an open-ended situation of identifying a problem to solve and eventually end with a possible solution that may be implemented.

Course Content:

CO 1: Designing attitude in day-to-day life:

Everybody design/innovate/ jugaad on almost daily basis, examples of everyday research, survey, observations, experimentation and problem solving

CO 2: What is the problem? How do people deal with it now? Key pain points?

Understanding user context (social, family, community, ecological, cultural, institutional, technological, political-economic etc.,)What, Why, How, for whom and Novelty (comparison with existing solutions)

CO 3: Ideation

Seeking solutions from existing designs, embedded thumb rules in people’s practices, nature’s patterns and strategies; constellation of technological, institutional, cultural and ecological endowments in which local practice is embedded

CO 4: Design thinking:

Design thinking including socio-ecological design/empathetic design, traditional wisdom based/ ancient wisdom based

CO 5: Paper design of the proposed solution

Review of the design by functional experts from Technical, Design, Social Sciences, Finance and Fabrication disciplines to give feedback to each team on their shortlisted ideas

Textbooks:

1) Ulrich, Karl T. and Steven D. Eppinger, Product Design and Development, 5th edition (2012), Irwin/McGraw-Hill

2) Sir Peter Medawar, .Advice to Young Scientist, Alfred P. Sloan Foundation Series, Edition 1^{st}

3) Santiago Ramón Caja, Advice to Young Investigator, The MIT Press, Edition 1^{st}

Websites:

1) National Innovation Foundation www.nif.org.in
Knowledge bank, http://summerschool.sristi.org/

Course Code : CML429
Course Title : Nanotechnology
Credits [LTP] :3 (3-0-0)
Pre-requisites : None
Overlap with : None

Objectives/Course Outcomes:
1) To understand knowledge of new technology
2) To understand the knowledge of Engg/Technology along with science
3) To understand the synthesis rout of Nano particles
4) To understand the applications in various Engg/Tech towards development of new product
5) To understand the approach for modeling for synthesis route

Program Outcomes for Chemical Engineering at VNIT:
a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

Course Objectives	Programme outcomes								
	A	b	c	d	e	f	g	h	i
CO1	√	√				√		√	√
CO2	√	√				√	√	√	
CO3	√	√							
CO4	√					√		√	√
CO5	√					√			

Content (CO Wise):
CO.1. To understand knowledge of new technology, Introduction to Nanotechnology, To understand the knowledge of Engg/Technology along with science, Physical chemistry of solid surfaces: Electrostatic stabilization, steric stabilization
CO.2. To understand the synthesis rout of Nano particles, Synthesis of Nanomaterials: Matrix mediated growth technique, sol-gel method, Chemical precipitation method etc,

CO.3. To understand the applications in various Engg/Tech towards development of new product

CO.4. Application in Chemical Technology: Polymer Nanocomposites-Synthesis, characterization, mechanical, thermal properties etc

CO.5. Application in Carbon nano tubes: Synthesis, characterization, SWNT, MWCNT, different models used for CNT, method of synthesis-Arc discharge method, lasor ablation method, CVD method etc

CO.6. Application in drug delivery, Application in nanofluids: Synthesis of various kind of nanofluids, application in Thermal conductivity, heat transfer coefficient, heat exchanger applications, study of dimensionless analogy, etc. To understand the approach for modeling for synthesis route. Study of Characterization of nanoparticles techniques: XRD, TEM, SEM, AFM, DSC, TGA, DMA, Rheometer etc

Text Books/ Reference Books

1. Gipzjpmg Cap, Ying Wang, Nanostructures and Nanomaterials (synthesis properties and applications, 2nd Edition, USA 2011
2. Jurgen Schulte, Nanotechnology (strategies, industry trends and applications, Willey,1st Edition, England 2005
3. S. Reich, C. Thomsen, J. Maultzsch, Carben Nanotubes (Basic concept and physical property, Wiley-VCH,1st Edition, 2004, Weinheim

Course Code : CML424
Course Title : Petroleum Refinery Engineering
Course Credits (L-T-P) : 3 credits (3-0-0)
Pre-requisites : Mass Transfer I
Overlaps with : None

Objectives:

1. To develop the fundamentals of refining of petroleum crude oil and its fractionation in different useful petroleum products.
2. The student will be aware to the product quality, related environmental concern and the standards by applying the different primary, secondary and advanced refinery processes.
3. The student will be aware to problems and remedies in petroleum sector.
4. To develop the skill and knowledge for upgradation of petroleum refineries as per present and future demand.
5. To develop the responsibility of technological inputs related to energy and environmental demand.

Outcomes:

At the end of the course, the student will understand the fundamentals and advances in refinery sector. Students will gain detailed knowledge of exploration of crude oil, its fractionation into different useful petroleum products, their quality, related environmental concerns and the standard by the recent and updated technology. The students will be able to visualize the scenario of refinery in India and abroad and can work in refineries and R&D sector of the related area.

Course description:

A brief review of the basic principles and existing techniques of petroleum refinery such as exploration of crude oil, characterization and fractionation into usable petroleum products. Recent advancements in secondary processes on the above areas to meet the revised standard and specification of the petroleum products. This course will end up with understanding of the fundamentals of refinery and present and future requirements of the refinery/oil sector.

Course Content:

*Unit I (6 hrs) :*Fundamental principles of origin and occurrence of petroleum crude and its exploration, Composition of petroleum, classification and physical properties, Characterization of crude oil and petroleum products, status of petroleum refining in India, future refining trends.
Unit II (6 hrs): Crude oil Distillation Process, Pretreatment of crude, atmospheric and vacuum distillation process
*Unit III (6 hrs) :*Secondary conversion processes: Thermal and catalytic cracking, Catalytic reforming, Pyrolysis
Unit IV (6 hrs): Heavy Residue Upgradation Technologies: Hydrocracking, Hydrotreating, visbreaking and coking, alkylation, Isomerisation, dehydrogenation processes, polymerization.

Unit V (6 hrs): Lubricating oil, grease and Bitumen: Dewaxing and deoiling, deasphalting, lube hydrofinishing, bitumen air blowing, Sweetening and Desulphurization, Hydrodesulphurisation of petroleum products.
Unit VI (6 hrs): Energy conservation in petroleum refineries. New Trends in petroleum refinery operations, Biorefinery concept.

Relationship of Course Objectives to Program Outcomes:

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	h	i
1	√	√	√				√	√	
2	√	√	√		√	√	√	√	√
3	√	√	√	√	√	√	√	√	√
4	√	√	√	√	√	√		√	
5					√	√	√	√	√

Textbooks:

1. Modern Petroleum Refining Processes, Bhaskara Rao B.K., Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi., Edition 3rd
2. Petroleum Refining Engineering, Nelson W.L.., Tata McGraw Hill Publication Co. Ltd. (1985), 4th Edition.
3. Petroleum Refining Technology and Economics, Gary J.H. &Handwerk G.E., Marcel Dekker, Inc., New York, 3rd Edition
4. Petroleum Refining Manual, Noel H.M., Publisher Reinhdd Pub. Corp., New York.

Modern Petroleum Technology, Hobson G.D. &Rohl W., Applied Science Publication, 4th Edition.

Course Code : **CML423**
Course Title : **Optimization Techniques**
Course Credit (L-T-P) : **3 Credits (3-0-0)**
Pre-requisites : **Basic Maths**
Overlapping/Equivalent courses : **None**

Course Objectives:

1) To understand the basics of optimization techniques, and problem formulation for optimization
2) To understand the single variable and multivariable optimization techniques and their application
3) To understand the linear programming application for optimization
4) To understand the advance optimization technique like genetic algorithm

Course Outcomes:

1) Student will understand necessary and sufficient condition for optimization and will be able to formulate the optimization problem.
2) Student will be able to solve different optimization problem and their application to the case studies like heat exchanger, evaporator etc

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	H	L				L		L	L
2	H	H				M			

Course description:

This course mainly deals with basics of different optimization techniques and its application for various engineering purpose.

Course Content:

CO1: Nature and organization of optimization problems: what optimization is all about, Why optimize, scope and hierarchy of optimization, examples of applications of optimization, the essential features of optimization problems, general procedure for solving optimization problems, obstacles to optimization. Classification of models, how to build a model, fitting functions to empirical data, the method of least squares, factorial experimental designs, fitting a model to data subject to constraints.

Basic concepts of optimization: Continuity of functions, unimodal versus Multimodel functions. Convex and Concave functions, Convex region, Necessary and sufficient conditions for an extremum of an unconstrained function, interpretation of the objective function in terms of its quadratic approximation.

Optimization of unconstrained functions: one-dimensional search:

Numerical methods for optimizing a function of one variable, scanning and bracketing procedures, Newton's, Quasi-Newton's and Secant methods of uni-dimensional search, region elimination methods, polynomial approximation methods, how the one- dimensional search is applied in a multi-dimensional problem, evaluation of uni-dimensional search methods.

CO2: Unconstrained multivariable optimization:

Direct methods, random search, grid search, uni-variate search, simplex method, conjugate search directions, Powell's method, indirect methods- first order, gradient method, conjugate method, indirect method- second order: Newton's method forcing the Hessain matrix to be positive definite, movement in the search direction, termination, summary of Newton's method, relation between conjugate gradient methods and Quasi-Newton method.

Linear programming and applications:

Basic concepts in linear programming, Degenerate LP's – graphical solution, natural occurrence of linear constraints, the simplex method of solving linear programming problems, standard LP form, obtaining a first feasible solution, the revised simplex method, sensitivity analysis, duality in linear programming, the Karmarkar algorithm, LP applications.

Optimization of Unit operations-1 recovery of waste heat, shell & tube heat exchangers, evaporator design, liquid liquid extraction process, optimal design of staged distillation column.

Optimization of Unit operations-2 Optimal pipe diameter, optimal residence time for maximum yield in an ideal isothermal batch reactor, chemostat, optimization of thermal cracker using liner programming.

Genetic Algorithms: (Qualitative treatment) Working principles, differences between GAs and traditional methods, similarities between GAs and traditional methods, GAs for constrained optimization, other GA operators, real coded GAs, Advanced Gas

Textbooks:

1) Edgar, T.F., D.M. Himmelblau, and L.S. Lasdon, Optimization of Chemical Processes, 2nd Edition, McGraw-Hill International Edition, Singapore, 2001.
2) Rao, S.S., Engineering Optimization Theory and Practice, 4th Edition, A Wiley Inetrscience Publication, Canada, 2009.
3) Kalyanmoy Deb, Optimization for Engineering Design, Algorithms and Examples, 2nd Edition, PHI Learning Private Limited, New Delhi, 2012

Reference Books:

1) Reklaitis, G.V., A. Ravindran, and K.M. Ragsdell, Engineering Optimization: Methods and Applications, 2nd Edition, John Wiley, New York, 2006.
2) Fletcher R., Practical method of optimization, 2nd Edition, John Wiley, New York, 2000.
3) Chong E.K.P. and Zal S. H., An Introduction to optimization, 2nd Edition, John Wiley, New York, 2001.
4) Nocedal J. and Wright S.J. Numerical Optimization, 2nd Edition, Springer,2000.
G. Mitsuo and C. Runwei, Genetic Algorithms and Engineering Optimization, John Wiley, New York, 2000

Course Code : CMD401
Course Title : Project Phase - I
Course Credits : 2 credits
Pre-requisites : None
Overlapping/Equivalent course : None

Course Content:

In this course, the candidate is expected to start his/ her basic preparation of experimental / mathematical project decided by the faculty advisor.

Course Outline: Molecular Simulation and Informatics

- **Course Code:** CML 402
- **Course Type:**Elective
- **Credits:** 03, Class Schedule: 3 Classes per week (55 minutes each)

I. Course Description

This course majorly focuses on providing the knowledge and experience of various molecular modeling, molecular simulation and informatics tools. The students will learn the concepts behind simulating the various types of molecular interactions and their utilization in design of new molecular systems.

II. Pre-requisites

- None

III. Course Objectives

The course aims:

1. To model and simulate themolecular interactions of various chemical systems.
2. To analyze themolecular information of various chemical systems for designing new molecules.

IV. Course Outcomes (COs)

Upon completing this course, students will be able to:

11. **CO1:** To build and apply molecular force field to chemical structures.
12. **CO2:** To understand and utilize molecular modeling, molecular simulation and data analysis tools to understand molecular level interactions
13. **CO3:** To understand and utilize molecular informatics tools for designing small molecular systems.

V. Expanded Course Description

Unit I: Introduction (6 Hours)

- Introduction and background of molecular computations
- Introduction to Windows and Linux based freeware molecular modeling, simulation and informatics tools

Unit II: ChemicalFile Formats (4 Hours)

- Chemical structure representation and manipulation
- SMILES specification rules and coding
- Various chemical file formats and their interconversion

Unit III: Molecular Modeling and Simulation (8 Hours)

- Applications of molecular computations in material design and development
- Molecular modeling,Molecular force field
- Fundamentals of molecular simulations

Unit IV: Molecular Simulations Concepts (8 Hours)

- Molecular dynamics simulations
- Integrating algorithms, Periodic box and Minimum image convention
- Long range forces and Non-bonded interactions
- Monte Carlo simulations and other advanced topics

Unit V: Simulation Data Analysis (6 Hours)

- Energy minimization, System relaxation
- Radial distribution function,
- Gromacs, Ambertools, Visual molecular dynamics
- Trajectory analysis

Unit VI: Virtual Screening Techniques (4 Hours)

- Virtual screening
- Ligand-based drug design: Quantitative structure activity relationship (QSAR) modeling
- Molecular descriptors, Lipinski's rule of five
- Structure-based drug design: Molecular docking, Search algorithms, Scoring functions

VI. Textbooks & References

1. Allen M. P. and Tildesley D.J., Computer Simulation of Liquids, 2nd Edition, Oxford University Press, 2017.
2. Frenkel D. and Smith B., Understanding Molecular Simulation: From Algorithms to Applications, 2nd Edition, Academic Press, San Diego, California, 2001.
3. Haile J. M., Molecular Dynamics Simulation: Elementary Methods, Wiley-Interscience, 1997.

4. Leach A. R., Molecular Modeling: Principles and Applications, Pearson, 2001, 2nd Edition.
5. Leach A. R. and Gillet V. J., An Introduction to Cheminformatics, Revised edition, Springer, 2007.

VII. Evaluation Scheme

- **Class Tests and Assignments:** 20%
- **Mid-Semester Exam (1.5 Hours):**25%
- **End-Semester Exam (3 Hours):** 55%

VIII. Relationship Between Course Outcomes (COs) and Program Outcomes (POs)

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	3	-	-	-	-	-	-	-
CO2	3	2	3	2	3	-	-	-	2	1	1	-
CO3	2	2	3	2	3	-	-	-	2	1	-	1

Key:

3 = Strong correlation, 2 = Moderate correlation, 1 = Weak correlation, - = No correlation

IX. Consultation Hours

Chamber consultation hours will be announced in class.

Course Code : CML 537
Course Title : Sustainable Engineering
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Objectives:

1) To introduce students to engineering design and manufacturing process including societal and environmental factors
2) To develop a fundamental understanding of the environmental impact of engineering decision/ design/system/ process/ product/ technology and method to assess
3) To develop a fundamental understanding of the societal impact of engineering decision/ design/system/ process/ product/ technology and method to assess

Course Outcomes:

1) Students will become aware of sustainability concept and issues related to it
2) Students will be able to measure sustainability index of a process
3) Sustainable design thinking will be developed
4) Students will be able to carry elementary LCA of a process

Program Outcomes for Chemical Engineering at VNIT:

- **a.** An ability to identify, formulates, and solves engineering problems
- **b.** An ability to design a system, component, or process to meet desired needs
- **c.** An ability to design and conduct experiments, as well as to analyze and interpret data
- **d.** An ability to communicate effectively
- **e.** An understanding of professional and ethical responsibility
- **f.** An ability to function on multidisciplinary teams
- **g.** The broad education necessary to understand the impact of engineering solutions in a global and societal context
- **h.** a recognition of the need for, and an ability to engage in lifelong learning
- **i.** knowledge of contemporary issues

Mapping with POs (Departmental reference):

Cos	Program Outcomes (PO)								
	a.	b.	c.	d.	e.	f.	g.	h.	i.
1.							H		H
2.		M			H		H		
3.		M		M	M	M	H		
4.		H		H	M	H	H	H	

Course description:

The course begins with exploration of the precept that we must design engineering systems for a world with a finite supply of natural resources and an earth with limited life support capacity. The course covers global policies and universal impact of sustainability. Course explores methods/ tools like optimization, process intensification, multi criteria analysis and multi criteria decision analysis for assessing sustainability (economic, environmental, societal impact). LCA analysis of technologies, process, process and product will be performed. Case studies from sections like energy, green engineering and bio refinery will be discussed.

Course Content (CO wise):

CO1: Introduction and definition:

What is Sustainability Engineering? various definitions of sustainability, Protocols regarding sustainable development, Basel convention, Cartagena protocol, Kyoto protocol, Stockholm convention, Measures of Sustainability

CO2: Tools for measuring Sustainibility

Optimization: Classical Optimization Techniques, Case Studies for Optimization, Process Intensification: Methods of Process Intensification, Multi Criteria Analysis (MCA): Criteria for Selecting MCA Techniques, Multi Criteria Decision Analysis

CO3: Life Cycle Analysis:

Why to Conduct a Life Cycle Assessment, definition and scope, Principles of life cycle analysis, Examples of LCA

CO4: Application Examples and case studies

Green chemistry/engineering, Energy: few case studies, Biorefineries

Textbooks:

1. Mary Ann Curran, Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products, Wiley, 2012, 1st Edition.
2. Leslie Jacquemin, Pierre Yves Pontalier, Caroline Sablayrolles *Life cycle assessment (LCA) applied to the process industry: a review* https://hal.archives-ouvertes.fr/hal-00741389
3. Alessio Ishizaka, Philippe Nemery, *"Multi Criteria Decision Analysis: methods and Software"*, Wiley-Blackwell, 2009

Reference Books:

1) Bhaskar D Kulkarni and Sachin A Mandavgane, Introduction to Sustainable Engineering: A course module for undergraduates, a course book for internal circulation among students.

Course Code : CML425
Course Title : Membrane Technology
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : Mass Transfer
Overlapping/Equivalent course : None

Course Outcomes/ Objectives:

1) To introduce the concept of various membrane separation processes such as microfiltration, reverse osmosis, ultrafiltration, nanofiltration, dialysis and electrodialysis, pervaporation, liquid membrane permeation, gas permeation
2) Identifying, formulating, and solving engineering problems involving various membrane separation processes
3) To designand analysis of various membrane separation processes such as microfiltration, reverse osmosis, ultrafiltration, nanofiltration, dialysis and electrodialysis, pervaporation etc.
4) To introduce application of various membrane separation processes such as microfiltration, reverse osmosis, ultrafiltration, nanofiltration etc.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with Pos:

POs → / COs ↓	a	B	C	d	e	f	g	h	i	j	k	l
CO1	H	L	L	L	L	H	M	L	M	M	L	L
CO2	H	H	M	L	L	M	L	L	L	M	L	L
CO3	H	H	H	L	L	M	L	L	L	L	L	M
CO4	M	L	M	L	L	H	M	L	M	L	M	M
Overall	H	H	M	L	L	M	M	L	M	M	L	M

Course content (CO wise):

CO1: Principles, characteristic, and classification of membrane separation processes; Membrane materials, structures, and preparation techniques; Membrane modules; Plant configurations.

CO2: Membrane characterization: Pore size and pore distribution; Bubble point test; Challenge test; Factors affecting retentivity, concentration polarization, gel polarization, fouling, cleaning and regeneration of membranes.

CO3: Mechanisms of separation: Porous membranes, dense membranes, and liquid membranes.Membrane separation models: Irreversible thermodynamics; Capillary flow theory; Solutiondiffusion model; Viscous flow models; Models for separation of gas (vapour) mixtures;

CO4: Science and technology of microfiltration, reverse osmosis, ultrafiltration, nanofiltration, dialysis and electrodialysis, pervaporation, liquid membrane permeation, gas permeation.Membrane reactors: Polymeric, ceramic, metal and bio-membrane

Textbooks/ Reference Books:

1) Geankoplis, Transport Processes And Separation Process principles, Prentice-Hall of India Private Ltd , New Delhi.

2) Richardson J.F., Harker J.H., Chemical Engineering, Vol. II, Butterworth Heinemann, New Delhi.2006.

3) Nath K., Membrane Separation Process, Prentice-Hall of India Private Ltd , New Delhi 2008

Course Code : CML426
Course Title **:** Polymer Processing
Course Credits : 3 credits (3-0-0)
Pre-requisites **:** None
Overlapping/ Equivalent courses **:** None

Course Objectives:

Based on the topics discussed in the class, Student shall be able to describe the principles of polymer processing techniques and evaluation of polymer properties as per standard.

Course Outcomes:

1) Differentiate and describe the processing technique employed for manufacturing plastic products.
2) Describe the flow behavior of polymers.
3) Identify the polymer by which the given product is made.
4) Describe the testing method for evaluation of specified property of a polymer.
5) Illustrate the importance of plastic materials and describe the plastic waste management techniques.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

Course Outcomes	**Program outcomes**								
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**I**
1	**H**			**H**			**H**	**M**	
2	**H**		**H**	**H**			**H**	**M**	
3	**H**		**H**	**H**	**L**	**H**	**H**	**H**	**M**

4	**H**		**H**	**H**			**H**	**H**	
5	**H**	**L**		**H**	**H**	**H**	**H**	**H**	**H**

Course description:

Basics on different types of polymers, structure-property relationship, different processing techniques for polymers, empirical rheological models, testing for various properties, management of polymer wastes and its recycling are stressed in this course

Course Content (CO wise):

CO1: Describe the Thermodynamic properties like free energy, activity, fugacity etc., for pure substances and solve the related problems.

Comparison of thermoplastics and thermoset plastics; Thermoset plastics - Types of resins, Interpenetrating Polymer Networks (IPN); Thermoplastics- Types of aliphatic and aromatic thermo plastics, copolymers, Blends and alloys; Liquid crystal plastics; cellular plastics; oriented plastic materials.

Basics of process design, Classification & general aspects of processes - molding& forming operations, Post die processing; Decoration of plastics - Printing, Vacuum Metalizing, In-molddecoration. Additives & Compounding - Different types of additives**,** Batch mixers, continuous mixers, Dispersive and distributive mixing, Characterization of mixed state.

CO2: Describe the Describe the flow behavior of polymers.

Fundamentals on Viscous & Viscoelastic behavior of polymer melt, Rheological measurements and Polymer processability. Non isothermal aspects - Temperature effect on rheological properties, Crystallization, Morphology & Orientation, plastic memory, Molecular weight effects on processing and properties.

CO 3: Identify the polymer by which the given product is made.

Basic concepts of testing, National& International standards, Test specimen preparation, Pre conditioning & Test atmosphere.

Identification of plastics by simple test - Visual examination, Density, Melting point, Solubility test, Flame test, Chemical tests.

CO4: Describe the testing method for evaluation of specified property of a polymer.

Effect of shape & structure on material properties, Long - term & short - term mechanical properties, crazing, Permeability & barrier properties, Environmental-stress cracking, Melt flow index, Heat deflection temperature, Vicat softening temperature, Glass transition temperature, thermal conductivity, Co-efficient of thermal expansion, Shrinkage, Thermal stability, Flammability.

CO5: Illustrate the importance of plastic materials and describe the plastic waste management techniques. Plastics waste and the associated problems, Integrated waste management - source reduction, recycling & sustainability correlation, energy recovering process. Environmental issues, policies and legislation in India.

Text Books:

1) Gruenwald G, Plastics - How Structure Determines Properties, Hanser Publishers, 1993
2) Vishu Shah, Hand Book of Plastics Testing Technology, John Wiley & Sons Inc.,New York, 2nd edition, 1998.

Reference Books:

1) Baird D. G. and Collias D. I., Polymer Processing Principles and Design, Butterworth-Heinemann, 1995
2) J.S.Anand, K.Ramamurthy, K.Palanivelu, How to identify Plastics by Simple Methods, CIPET, Chennai, 2nd edition.
3) Anthony L. Andrady (Ed.), Plastics and the Environment, Wiley Interscience, New York.

Course Code : CML427
Course Title : Advanced Separation Processes
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Brief Description: This course will cover the advanced separation methods used in chemical and other processes. It will focus on understanding the need for advanced separation methods, its principles and basic design calculations. Further, the selection of appropriate separation method will also be covered

Course Objectives:

- To explore the principles, design, and application of membrane-based separation techniques such as microfiltration, ultrafiltration, nanofiltration, and reverse osmosis.
- To examine a wide range of advanced separation methods, such as, Surfactant based separation methods, ATPS, chromatography, isoelectric focusing, salting out etc and its applications
- To focus on the specific applications of advanced separation methods in biotechnology, bioprocessing, and chemical engineering, including cell harvesting, protein purification, and product isolation.
- To Investigate emerging separation techniques

Course Outcomes:
CO1: Ability to understand the governing mechanism and driving force of various advanced separation processes
CO2: Ability to perform process and design calculations on membrane separation processes Ability to understand process and design consideration on chromatography, aqueous two phase separations, bio separations
CO3: Ability to understand process and design consideration on hybrid and reactive separations
CO4: Ability to select/choose the suitable separation method for a particular product

Mapping of CO and PO:

CO/PO	PO											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								1
CO 2	3	3	3	1		2	2					
CO 3	3	3	3	2	3	3	3	3			2	2
CO 4	3	3	3	3	3	3	3	3	2			3

Expanded Course Description:

Overview of Separation Processes: Introduction / Revision of various Conventional Separation Processes and their applications, advantages, and disadvantages, Need of advance Separation processes, types.

Membrane Separations: Introduction, type of membrane separations, membrane, membrane materials, ultrafiltration, microfiltration, nanofiltration, reverse osmosis, pervaporation, type of membrane modules, membrane fouling, concentration polarization, various mathematical models for membrane processes, application, design considerations.

Chromatographic Separation: Introduction, Principles, Classifications, High performance liquid chromatography, ion exchange chromatography, affinity chromatography, reversed phase chromatography, gas chromatography, application. Bio-separation Processes: Introduction, overview of bioseparations, cell disruption/ pretreatment methods, filtration, centrifugation, adsorption, extraction / aqueous two phase systems, membrane separation, precipitation, chromatographic separation, Applications. Isoelectric focusing, Various case studies

Hybrid Separations: Introduction, Concept of hybrid separations, types of hybrid separations, networking or combination of various separation processes, applications, design aspects, scope for future.

Reactive Separations: Introduction, Concept of reactive separations, types of reactive separations, reactive distillation, membrane based reactive separations, reactive extraction, reactive adsorption, reactive absorption, reactive crystallization, applications, design aspects, scope for future.

Textbooks/References:

1) Seader J. D., Henley E. J., Separation Process Principles, Wiley, 2001, 2nd Edition

2) Richardson J. F., Harker J. H., Chemical Engineering Vol. 2, Elsevier, 2002, 5th Edition.

3) Mukhopadhyay M., Natural Extract using Supercritical CO_2, CRC Press, 2000, 1st Edition.

4) Nath K., Membrane Separation Processes, Prentice Hall of India, 2008, 1st Edition.

5) Sivasankar B., Bio-separations: Principles and Techniques, Prentice Hall of India, 2005, 1st Edition.

Course Evaluation:

- Class Tests and Assignments: 20%
- Mid-Semester Exam (1.5 Hours): 25-30%
- End-Semester Exam (3 Hours): 50-55%

Course Code : CML428
Course Title **:** CFD for Chemical Engineers
Course Credits (L-T-P) :3 credits (3-0-0)
Pre-requisites **:** Fluid Mechanics
Overlapping/Equivalent : None

Course Outcomes/ Objectives:

1) Students shall be able to apply basic principles used in CFD.
2) Able to formulate the problem and apply discretisation techniques.
3) Able to design and solve heat transfer problems using CFD techniques.
4) Able to design and solve fluid flow problems using CFD techniques.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. a recognition of the need for, and an ability to engage in lifelong learning
i. knowledge of contemporary issues

Mapping with POs (Departmental reference) *:

POs → COs ↓	a	B	C	D	e	f	G	h	i	j	k
CO1	**H**	**L**									
CO2	**H**	**H**		**L**	**L**						
CO3	**H**	**H**	**H**	**H**	**M**	**M**		**M**	**M**		
CO4	**H**	**H**	**H**	**H**	**M**	**M**		**M**	**M**		
Overall	**H**	**H**	**M**	**M**	**M**	**M**		**M**	**M**		

CourseContent (CO wise):

CO1: *Introduction*

History, Comparison of the three basic approaches for engineering problems in solving by analytical, experimental and computational methods, Beam advance in computational techniques, Softwares available for CFD.

CO2: *Problem Formulation*

Formulation of problem, Physical and mathematical classification of problems, Types of governing differential equations.

CO3: *Discretisation*

Truncation and Round-off error; Explicit and Implicit approaches; Basic of finite difference method, Finite element method, Finite volume method and Spectral method, Treatment of boundary conditions, Examples using Comsol Multiphysics Software on Dirichlet boundary condition, Neumann boundary condition, etc.

CO4: *Numerical Solution of Heat Conduction Problems*

Steady-state problems, One dimensional heat conduction transfer through a pin-fin, Two dimensional conduction through a plate unsteady state problem, One dimensional transient heat conduction, Explicit and implicit methods, Assessing accuracy and stability of numerical methods, Examples using Comsol Multiphysics Software on state heat transfer in one dimension, transient heat transfer in one dimension, heat transfer in two dimensions, heat conduction with a hole, etc.

CO5: *Numerical Solution of Fluid Flow Problems*

Types of fluid flow and their governing equation, Viscous incompressible flows calculation of flow field using the stream function-vorticity method, Calculation of boundary layer over a flat plate, Numerical algorithm for solving complete Navier-Stokes equation-MAC method SIMPLE algorithm, Introduction to standard κ-ε model for turbulent incompressible flow, Project problem, Examples using Comsol Multiphysics Software on flow of newtonian fluid in a pipe, flow of non-newtonian fluid in a pipe, reaction and diffusion, turbulent flow in a pipe, determining arrhenius parametrs using parameter estimation, etc.

Textbooks:

1. P. S. Ghoshdastidar, Computer simulation of flow and heat transfer, Tata McGraw-Hill Publishing, 1st Edition, 1998.
2. K. Muralidhar and T. Sundararajan, Computational fluid flow and heat transfer, Narosa Publications, 2nd Edition, 2003.

Reference Books:

1) H. K. Versteeg and W. Malalasekera, An introduction to CFD, Longman Scientific and Technical, 1st Edition, 1995.

2) J. H. Ferriger, M. Peric, Springer, Computational methods for fluid dynamics, 1st Edition, 1996.

3) S. V. Patankar, Numerical heat transfer and fluid flow, Mc Graw-Hill Book Company, 1st Edition, 1980.

1. Finlayson, B A, Introduction to Chemical Engineering Computing, Wiley, 2nd Edition, 2014.

Course Code	**: CML481**
Course Title	**: Instrumental Analytical Techniques**
Course Type	**: Core**
Course Credits (L-T-P)	**: 3 (3-0-0)**
Pre-requisites	**: CML221: Mass Transfer-I, CML222: Heat Transfer**
Overlaps with	**: None**

Course description:
This course mainly deals with basics and application of various analytical equipments like, gas chromatography, liquid chromatography, gas chromatography mass spectroscopy, liquid chromatography mass spectroscopy, UV visible Spectrophotometer and infra red spectrophotometer etc.

Objectives:
Students will be able to identify and differentiate the analytical equipments for chemical analysis as per requirement

Course Outcomes:
1) Describe the various important analytical equipments and its fundaments principle and functioning.
2) Using the output information of different analytical equipments, for corresponding chemical analysis the heat mass and momentum transfer

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	L	L	H		L				
2	M	M	H		M				

Program Outcomes for Chemical Engineering at VNIT:
a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a
h. global and societal context
i. A recognition of the need for, and an ability to engage in lifelong learning
j. A knowledge of contemporary issues

Course Content:

CO 1: An introduction to analytical chemistry: choice of analytical methodology, sampling, sample preparation, chemical analysis, tools for quantitative chemical analysis, quality assurance. Extraction methods such as liquid-liquid extraction, solid phase extraction, super-critical fluid extraction and accelerated solvent extraction. Cleanup and fractionation methods.

CO 2: Introduction to Chromatography, high-pressure liquid chromatography (HPLC), gas chromatography (GC) and other chromatographic methods. Detector types with focus on mass spectrometry and hyphenated techniques such as GC-MS and LC-MS.

Introduction to spectroscopic methods (UV-VIS, IR, X-ray, atomic absorption spectroscopy (AAS) and inductive coupled plasma mass spectrometry).

Introduction to data processing, errors in chemical analyses, statistical analyses (including chemometrics) and data presentation. Method development, evaluation, validation and QA/QC measures. Uncertainty analysis.

Textbooks:

1) Harris, D.C., Quantitative Chemical Analysis, 7th Edition, W.H.Freeman and company, New York 2006.

2) Bruno, T.J, and Svoronos, P. D. N., Handbook of Basic Tables for Chemical Analysis, 2nd Edition, CRC Press, New York 2003.

Reference Books:

1) McNair, H. M. and Miller, J. M., Basic Gas Chromatography, 1st Edition, John Willy and Sons, Inc, Singapore, 1998.

2) Palvia D. L., Lampman G. M., Kriz G. S. and Vyvyan J. R., Introduction to Spectroscopy, 4th Edition, Brooks/Cole, Belmont USA, 2009.

Snyder L. R, and Krikland J. J., Introduction to Modern Liquid Chromatography, 2nd Edition, A Wiley Inetrscience Publication, New York, 1979. C.O. Bennet and J.E. Myres, Momentum, Heat & Mass Transfer, McGraw Hills, 3rd Editon, 1994.

Course Code : CML529
Course Title : Optimization Techniques in Process Design
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : Basic Maths
Overlapping/Equivalent courses : None

Course Objectives:

1) To understand the basics of optimization techniques, and problem formulation for optimization
2) To understand the single variable and multivariable optimization techniques and their application
3) To understand the linear programming application for optimization
4) To understand the advance optimization technique like genetic algorithm

Course Outcomes:

1) Student will understand necessary and sufficient condition for optimization and will be able to formulate the optimization problem.
2) Student will be able to solve different optimization problem and their application to the case studies like heat exchanger, evaporator etc

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with Pos (Departmental references):

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
CO1	H	L				L		L	L
CO2	H	H				M			

Course description:

This course mainly deals with basics of different optimization techniques and its application for various engineering purpose.

Course Content (CO wise):

CO1: Nature and organization of optimization problems: what optimization is all about, Why optimize, scope and hierarchy of optimization, examples of applications of optimization, the essential features of optimization problems, general procedure for solving optimization problems, obstacles to optimization. Classification of models, how to build a model, fitting functions to empirical data, the method of least squares, factorial experimental designs, fitting a model to data subject to constraints.

Basic concepts of optimization: Continuity of functions, unimodal versus Multimodel functions. Convex and Concave functions, Convex region, Necessary and sufficient conditions for an extremum of an unconstrained function, interpretation of the objective function in terms of its quadratic approximation.

Optimization of unconstrained functions: one-dimensional search:

Numerical methods for optimizing a function of one variable, scanning and bracketing procedures, Newton's, Quasi-Newton's and Secant methods of uni-dimensional search, region elimination methods, polynomial approximation methods, how the one- dimensional search is applied in a multi-dimensional problem, evaluation of uni-dimensional search methods.

CO2: Unconstrained multivariable optimization:

Direct methods, random search, grid search, uni-variate search, simplex method, conjugate search directions, Powell's method, indirect methods- first order, gradient method, conjugate method, indirect method- second order: Newton's method forcing the Hessain matrix to be positive definite, movement in the search direction, termination, summary of Newton's method, relation between conjugate gradient methods and Quasi-Newton method.

Linear programming and applications:

Basic concepts in linear programming, Degenerate LP's – graphical solution, natural occurrence of linear constraints, the simplex method of solving linear programming problems, standard LP form, obtaining a first feasible solution, the revised simplex method, sensitivity analysis, duality in linear programming, the Karmarkar algorithm, LP applications.

Optimization of Unit operations-1 recovery of waste heat, shell & tube heat exchangers, evaporator design, liquid liquid extraction process, optimal design of staged distillation column.

Optimization of Unit operations-2 Optimal pipe diameter, optimal residence time for maximum yield in an ideal isothermal batch reactor, chemostat, optimization of thermal cracker using liner programming.

Genetic Algorithms: (Qualitative treatment) Working principles, differences between GAs and traditional methods, similarities between GAs and traditional methods, GAs for constrained optimization, other GA operators, real coded GAs, Advanced Gas

Textbooks:

1) Edgar, T.F., D.M. Himmelblau, and L.S. Lasdon, Optimization of Chemical Processes, 2nd Edition, McGraw-Hill International Edition, Singapore, 2001.
2) Rao, S.S., Engineering Optimization Theory and Practice, 4^{th} Edition, A Wiley Inetrscience Publication, Canada, 2009.

3) Kalyanmoy Deb, Optimization for Engineering Design, Algorithms and Examples, 2nd Edition, PHI Learning Private Limited, New Delhi, 2012

Reference Books:

1) Reklaitis, G.V., A. Ravindran, and K.M. Ragsdell, Engineering Optimization: Methods and Applications, 2nd Edition, John Wiley, New York, 2006.
2) Fletcher R., Practical method of optimization, 2nd Edition, John Wiley, New York, 2000.
3) Chong E.K.P. and Zal S. H., An Introduction to optimization, 2nd Edition, John Wiley, New York, 2001.
4) Nocedal J. and Wright S.J. Numerical Optimization, 2nd Edition, Springer,2000.
5) G. Mitsuo and C. Runwei, Genetic Algorithms and Engineering Optimization, John Wiley, New York, 2000.

Course Code	: CMD402
Course Title	: Project Phase -II
Course Credits	: 4 credits
Pre-requisites	: None
Overlapping/Equivalent course	: None

Course Content:

In this course, the candidate is expected to start his/ her basic preparation of experimental / mathematical project decided by the faculty advisor.

Course Code : CML386
Course Title : Biotechnology & Biochemical Engineering
Course Credits (L-T-P) : 03 Credits (3-0-0)
Pre-requisites : Chemical Reaction Engineering
Overlapping/Equivalent courses : None

Course Objectives:

To encourages the students to work in the field of bio-chemical Industry. The objective of this program is to introduce the basic concepts of bio-processing to the chemical engineers.

Course Outcomes:

1) Student will get well acquainted with intervention of chemical engineering principles to biological process
2) Student will able to understand the concepts of enzyme technology, immobilization technology, cell growth kinetics.
3) Student will get knowledge of bioreactors, fermentation, sterilization techniques, and media preparation.
4) Student will get well acquainted with biological process specific reactor design aspects.
5) Student will get knowledge of downstream processing in fermentation industries.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. Recognition of the need for, and an ability to engage in lifelong learning.
i. A knowledge of contemporary issues

Mapping with Pos (Departmental reference)

Course Objectives	Programme outcomes								
	a	b	c	d	e	f	g	H	i
CO1	H								
CO2	H	H					L	M	
CO3	H	H					L	M	
CO4									
CO5	H	M				L			

Course description:

A consideration of the engineering and scientific basis for using cells or their components in engineered systems. Central topics addressed include kinetics and reactor design for enzyme and cellular systems; fundamentals, techniques, and bio-separations. Additional lectures will provide an introduction to metabolic modeling as well as special topics. The course is designed to be accessible to students with engineering backgrounds.

Course Content:

Types of micro-organisms, structure and function of microbial cells, batch and continuous culture, microbial growth kinetics, enzymes from cells, their function and immobilized kinetics, kinetics of microbial growth. Enzyme technology and kinetics, enzyme catalysis, enzyme applications in industries and medicines, metabolism and bioenergetics, photosynthesis, synthesis and regulation of bimolecular, fundamentals of microgenetics, role of DNA and RNA. Reactions catalyzed by enzymes, types of reactors such as CFSTR, Plug flow. Sterilization techniques, media and air sterilization, death rate of microorganisms. Introduction to fermentor design, design of fermentors with modified organisms. Introduction to Bioreactor design, scale up of bioreactions and bioreactors, volumetric mass transfer rate of oxygen from air bubbles, respiratory model for mycellial pallet, mechanical mixing, aeration, power consumption, heat transfer in bio reactor. Bioreactor modeling and simulation. Design for bioproducts, applications in biochemical and biomedical engineering.

Downstream processing in biochemical industries: such as separation processes for bulk chemicals unit operations such as Ultra filtration, Aqueous two-phase extraction.

Textbooks:

1) Bioprocess Engineering: -Basic concept by Shuler &Kargi (PHI)

Reference Books:

1) Biochemical Engineering fundamentals By Bailey ollis

2) Biochemical Engineering: -principles & concepts by Syed Tanveer Ahmed Inamdar(PHI)

3) Introduction to Biochemical Engineering by D.G.Rao

Course Code : CML422
Course Title : Plant Utility
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Objectives:

Student will be able to describe the different utilities used to run the process plant. Acquire the knowledge for selection of different utilities. Understand basic calculation involved in steam generation, psychometric operation, cooling tower and refrigeration.

Course Outcomes:

1) State the principles involved during water treatment, generation of steam and its uses, refrigeration cycles.
2) Describe the different equipments used to run the process plant with different utilities.
3) Acquire the knowledge for selection of different utilities.
4) Understand basic calculation involved in steam generation, psychometric operation and refrigeration

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in aglobal and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	Program outcomes								
	a	B	c	d	e	f	g	h	i
CO1	**H**			**H**			**H**	**M**	
CO2	**H**		**H**	**H**			**H**	**M**	
CO3	**H**		**H**	**H**	**L**	**H**	**H**	**H**	**M**
CO4	**H**		**H**	**H**			**H**	**H**	

Course Content (CO wise):

The course covers the major utilities required for process plants such as water and its treatment, properties of steam and boiler performance, different refrigerants and refrigeration cycle, air compressor and psychometric properties. It also involves the basic calculations for evaluating the performance of steam generation, refrigeration, compressor and cooling tower

CO1: Importance of utilities: Sources of water, hard and soft water , Requisites of industrial water and its uses, Methods of water treatment, Chemical softening, Demineralization, Resins used for water softening, Reverse osmosis and membrane separation, Effects of impure boiler feed water & its treatments., Scale & sludge formation, Corrosion, Priming & foaming, Caustic embrittlement.

CO2: Refrigeration: Refrigeration cycles 04, Different methods of refrigeration used in industry, Vapour compression, Vapour absorption: Lithium bromide (eco-Friendly)‘ Different refrigerants‘ Monochlorodifluoro methane (R-22)‘ Chlorofluorocarbons (CFC-Free) ‘ Secondary refrigerants: Brines’ Simple calculation of C.O.P. Refrigerating effects.

CO3: Steam and steam generation: Properties of steam, Problems based on enthalpy calculation for wet, steam, dry saturated steam, superheated steam,, Types of steam generator / boilers: water tube & fire tube, Solid fuel fired boiler. , waste gas fired boiler., Waste heat boiler., Fluidized bed boiler., Scaling, trouble shooting, preparing boiler for inspection, Steam traps, boiler mountings and accessories , Boiler Act.

CO4: Psychrometry: Properties of Air-water vapors. Use of humidity chart , Equipment used for humidification, dehumidification, Evaporative cooling, spray ponds, cooling towers . Air: Use of Compressed air, process air and instrument air, Process of getting instrument air. Non steam heating system, Thermic fluid heater, Down thermo heater, Temperature range.

Text Books:

1) Chattopadhya Boiler operations Tata McGraw Hill, New Delhi
2) Yadav R. Thermodynamics & Heat Engines Central Publishing House
3) Lyle O. Efficient Use of Steam Prentice Hall 1963

Reference Books:

1) Mahesh Rathore Thermal Engineering McGraw Hill

Course Code : CML431
Course Title : Entrepreneurship Development
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Outcomes/Objectives:

1) To understand the role of an entrepreneur.
2) To understand the factors influencing an entrepreneur.
3) To understand the steps involved in setting up a business.
4) To understand the financial aspects of a business.
5) To understand the support given to entrepreneurs.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with POs (Departmental reference):

Course Objectives	**Programme outcomes**								
	A	**b**	**c**	**d**	**E**	**f**	**g**	**H**	**i**
CO1					**H**	**M**		**M**	
CO2									
CO3	**M**	**M**				**M**	**M**	**M**	**M**
CO4	**M**	**M**				**M**	**L**	**L**	**M**
CO5								**M**	**H**

Course Content (CO wise):

CO1: To understand the role of an entrepreneur.

Entrepreneur: Traits of Entrepreneurs - Types of Entrepreneurs – Intrapreneur, Difference between Entrepreneur and Intrapreneur - Entrepreneurship in Economic Growth, Factors affecting Entrepreneurial Growth.

CO2: To understand the factors influencing an entrepreneur.

Motivation: Major motives influencing Entrepreneur- Achievement Motivation Training, Self Rating, Business game, Thematic Apperception Test - Stress Management. Entrepreneurship Development Programs - Need, objectives.

CO3: To understand the steps involved in setting up a business.

Business: Small Enterprises-definition, Classification - Characteristics, ownership structure Project Formulation – Steps involved in setting up a Business - Identifying, Selecting a good business opportunity Market survey and Research, Techno economic Feasibility Assessment - Preliminary Project Report-Project Appraisal-Sources of information-Classification of needs and Agencies.

CO4: To understand the financial aspects of a business.

Financing & Accounting: Need-Sources of Finance, Term Loans, Capital structure, Financial Institutions, Management of working capital, Costing Break Even Analysis, Network analysis Techniques of PERT/CPM - Taxation - Income Tax, Excise Duty - Sales Tax.

CO5: To understand the support given to entrepreneurs.

Support to Entrepreneurs: Institutional Support to Entrepreneurs-Sickness in small Business - Concept, Magnitude, Causes and Consequences, Corrective measures - Government Policy for small Scale Enterprise - Growth strategies in small Industry - Expansion, Diversification, Joint venture, Merger, sub-contracting.

Text Books:

1) Dr Sharma D. D., Total Quality Management, Sultan Chand and Sons, New Delhi

2) Gupta C. B. and Srinivasan P., Entrepreneurship Development, Sultan Chand and Sons, New Delhi

Reference Books:

1) Khanna S.S, Entrepreneurial Development, S.Chand& Co. Ltd NewDelhi, 1999.

2) Philip Kotler, Marketing Management, Prentice Hall of India, New Delhi.

3) Rathore B.S. and Dr. Saini J. S, A Handbook of Entreprenuership, AapgaPublications, Panchkula (Haryana).

Course Code : CML468
Course Title : Ore and Mineral Processing
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course Objectives:

1) The objective of this course is to understand the fundamentals of minerals processing
2) To give the basic principles of different unit operation used in mineral dressing
3) To identify various processes and equipment used in mineral processing.

Course Outcomes:

At the completion of the course students will be able to understand the different steps used for the processing of various minerals.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility.
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with Pos (Departmental reference):

Course Outcomes	Program outcomes								
	a	b	c	d	e	f	g	h	i
1	H			H			H	M	
2	H		H	H			H	M	
3	H		H	H	L	H	H	H	M

Course Content (CO wise):

The course covers the fundamentals of mineral processing , aspects of sampling, particle characterization, comminution and classification, physical processes of mineral concentration involving dense media and gravity separation, magnetic and electrostatic separation and ore sorting techniques, Froth flotation technique etc.

CO1: Mineralogy: Studies of important metallic and non metallic minerals, their characteristics, origin etc. application of non metallic minerals. Sea as a source of minerals. Status of mineral beneficiation industry in India. Study of some representative beneficiation practices with flow sheets. Sampling methodology and equipments.

CO2: Comminution: Primary, secondary and special crushers and their performances. Cylindrical and cylindro-conical ball mills, Rod mills, Tube / Pot mills and their performances, capacities, reduction ratios etc. Dry and Wet Grinding. Open and closed circuit grinding. Work Index calculations. Interlocking and liberation of minerals.

CO3: Screening: Sizing and Classification: Standard screening tests and graphical representations of the results. Particle size distribution, Sorting, Sizingand Pneumatic classifiers and their performances.Thickeners, Hydrocyclonesetc.Theory and practice of sedimentation and filtration. Working of Rotary vacuum filters. Gravity Concentration Techniques: Principles of Jigging, Tabling and Heavy Media Separation. Processes with equipments used, important controlling factors in operation and application. Beneficiation practice for arsenopyrite containing scheelite. Froth Flotation: Natural and Artificial Floatability of minerals. Frothers, Collectors, Depressants, Activators/Deactivators, PH Modifiers, etc. Flotation machines.Study of representative sulphide and non sulphide minerals and non metallicores.Multistage flotation and Column Flotation. Electrostatic and Magnetic Separation: Principles of Electrostatic and Magnetic Separation (Dry and Wet type). Separation units used in practices and examples in the industries. Calculation of Recovery and ratio of concentration and Mass balance calculations in ore dressing. Industrial set up of Ore Dressing plant.

Text Books:

1) Mineral Processing Author Pryor E.J; Publisher Kluwar Academic Publishers Edition 3rd Edition

2) Elements of Mineralogy Author Rutley F. Publisher Thomas Murray & Co., London

3) A Text Book of Ore Dressing, Author Robert H. & Locke, Richards C.E; Publisher McGraw Hill Co.

Reference Books:

Elements of Mineral Dressing Author Gaudin A.M.;, Publisher New York Edition2nd Edition

Course Code : CML530
Course Title : PROCESS INTENSIFICATION
Course Credits (L-T-P) : 3 Credits (3-0-0)
Pre-requisites : None
Overlapping/Equivalent courses : None

Course description:

The course presents the concept, philosophy, types and application of process intensification from a chemical engineering viewpoint. Various ways of intensification and its application are stressed in this course.

Objectives:

Based on the concepts taught, Student shall be able to apply the concepts and philosophy of process intensification to resolve the various issues in process industry such as cost, waste minimization, size etc.

Outcomes:

1. Describe the basic philosophy and concepts of process intensification
2. Describe various methods of process intensification.
3. Describe various equipments of process intensification.
4. Describe the application of process intensification.

Mapping Course Outcomes to Program Outcomes:

Course Outcomes	**Program outcomes**								
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**
1	**H**			**H**				**L**	
2	**H**	**L**	**M**				**H**	**M**	
3	**H**	**L**	**M**				**H**	**M**	
4	**H**	**M**	**L**			**M**	**H**	**M**	**H**

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams

g. The broad education necessary to understand the impact of engineering solutions in a

h. global and societal context

i. A recognition of the need for, and an ability to engage in lifelong learning

j. A knowledge of contemporary issues

Course Content:

CO 1: Introduction to Process Intensification: History, Philosophy, principles.

CO 2: Reactive and hybrid separations: concept of reactive separations, reactive distillation, membrane based reactive separation reactive adsorption, reactive extraction, reactive crystallization, hybrid separations, extractive distillation, adsorptive distillation, , membrane distillation , membrane chromatographic separation, design, applications. High Gravity Fields: process fundamentals, Rotating packed Bed, Design, Applications, Scale-up Spinning Disc Reactor: mathematical models, heat & mass transfer, design, application. Compact multifunctional heat exchangers: types, applications, design. Multifunctional reactors: Concept, integration of reaction, mass and heat transfer, design, application, various equipments, Process synthesis and integration: conventional design, conceptual design, elements, reaction engineering, complex distillation, systems, industrial studies.

CO 3: Inline and high intensity mixers: concept of mixing, motionless mixers, mixing and reaction, design, gas-liquid mixing, combined heat exchanger and reactors, design, applications. Microreaction technology: enhancement of heat and mass transfer, control and safety, fabrication, application, design. Structured catalyst and reactors: monolithic reactors, catalysts, gas-phase reactions, application, design

CO 4: Process intensification for industrial safety: concept of industrial safety, hazardous reactions and safety, applications, examples. Industrial practice: methodology and applications, commercial examples of process intensification

Textbooks:

1. Re-engineering the chemical processing plant: Process Intensification, Andrzej Stankiewicz, JacobA. Moulijn, Marcel Dekker, Inc., Marcel Dekker, Inc., 1st edition, 2004

Reference Books:

Process Intensification in Practice, Cornelis de Weerd, John Wiley and Sons, 1st edition, 2005

Course Code : **CML 342**
Course Title : **Safety and Risk Analysis**
Course Credits (L-T-P) : **3 Credits (3-0-0)**
Pre-requisites : **None**
Overlapping/Equipment courses : **None**

Course Outcomes/Objectives:

1) To give students the knowledge on safety programs, engineering ethics, plant safety measures, safety warning symbols, handling of hazardous chemicals, fire andexplosion hazards and different methods of hazard identification and its analysisin qualitative and quantitative scales.

2) To gain knowledge on different mathematic models used to predict the rate of leakage of fuels from various unit operations.

3) To gain knowledge on the design to prevent fires and explosion, risk assessment and case study problems.

4) To develop the safety concepts among the students with detailed understanding of technical knowledge.

5) To develop the responsibility and ability for precautions and remedial actions for any untoward event.

Program Outcomes for Chemical Engineering at VNIT:

a. An ability to identify, formulates, and solves engineering problems
b. An ability to design a system, component, or process to meet desired needs
c. An ability to design and conduct experiments, as well as to analyze and interpret data
d. An ability to communicate effectively
e. An understanding of professional and ethical responsibility
f. An ability to function on multidisciplinary teams
g. The broad education necessary to understand the impact of engineering solutions in a global and societal context
h. A recognition of the need for, and an ability to engage in lifelong learning
i. A knowledge of contemporary issues

Mapping with Pos (Departmental reference)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO-1	H					M		H		L		H
CO-2	H	H	M									H
CO-3	H	L	H									H
CO-4	M	M	L		H							H
CO-5	M	H			M				H	M		H

Course description:

The course deals with the safety measures, risks, hazards, designs to prevent fires and explosions in chemical process industries and the analysis of event. The course will cover the important technical fundamentals of chemical process safety, risks and hazard assessment and their analytical valuation for prevention and preparation of standards for future.

Unit-I: Introduction to process plant safety, handling of hazardous chemicals, OSHA standards, importance of plant layout in safety, importance of site selection, personnel safety, role of human error in losses. Case studies of fires, explosions, disasters in chemical process plants.

Unit-II: Lower flammability limit (LFL), UFL, LEL, UEL, TLV, electrostatic hazards, Hazard code and explosive limit, TWA, Ceiling level, Safety in handling of gases, liquids and solids, Flammable liquid hazards, fire and explosion index, fire ball hazards, oil spillage hazards, Bleveuvce, pool fires, jet fires, radiation hazards. Explosion, emergency and disasters in chemical process plants, onsite and offsite emergency plan, Fire detectors, smoke detectors. Resources for combating fires, dry chemical powders, fire fighting foam, fixed and portable fire extinguishers, FMEA.

Unit-III: Designs to Prevent Fires and Explosions- Vacuum Purging, Pressure Purging, Combined Pressure-Vacuum Purging, Vacuum and Pressure Purging with Impure Nitrogen, Advantages and Disadvantages of the Various Pressure and Vacuum inerting procedures, Sweep-Through Purging, Siphon Purging.

Unit-IV: Introduction to Source Models, Flow of Liquid through a Hole, Flow of Liquid through a Hole in a Tank, Flow of Vapor through Holes, Flashing Liquids, Flow of Gases through Pipes-Adiabatic Flows and isothermal flows, Liquid Pool Evaporation or Boiling.

Unit-V: Safely audit of chemical process plants, HAZOP studies, RiskAssesment- fault tree and event tree analysis. Risk analysis of chemical processes, risk management, risk identification, personnel training, risk to environment.

Course Content (CO wise):

At the end of the course, the student Students will gain knowledge on

CO1: Basic procedures to be followed to ensure safety in chemical process industries

CO2: Fire-flammability characteristics of vapors and liquids

CO3: Designs to prevent fires and explosions

CO4: Mathematical models to understand the rate of discharge of materials

CO5: Risk assessment of various chemical process plants

Textbooks:

1) Chemical Process Safety: Fundamentals with Applications, Daniel A. Crowl and Joseph F. Louvar, Prentice Hall International Series, 2nd Edition.

2) Safe and Efficient Plant Operation and Maintenance, Greene R., McGraw Hill Book Co., New York.

3) Safety Management and Practices for Hazardous Units, Dekkar Marcel, McGraw Hill Book Co., New York, 1995.

Reference Books:

1) Safety and Good House Keeping, Saxena, National Productivity Council, New Delhi (1976), 3rd Edition.

Safety in Process Plant Design, Wells G.L., George Godwin Ltd., (1980).